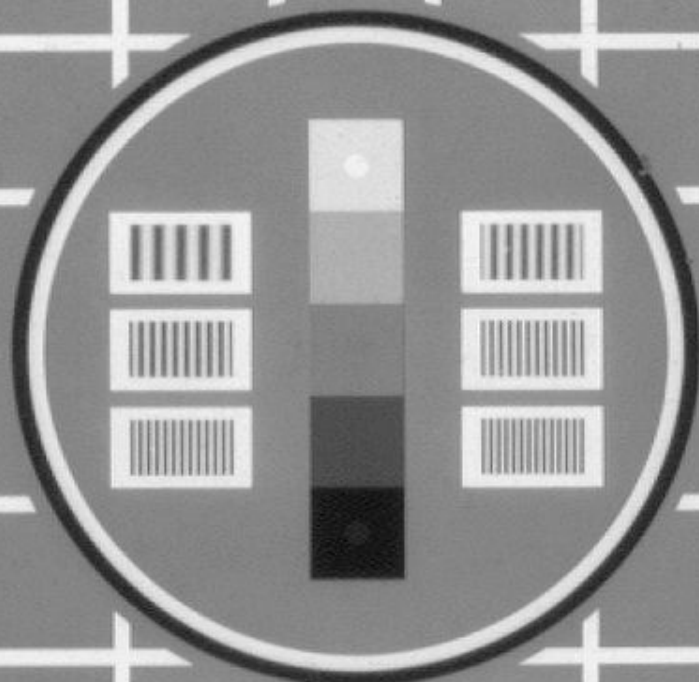


Broadcast Technology UG1

TV Standards, Colour Space



• | D | •

BBC 1 405 LINES

TV Broadcast Standards *(some of ...)*

Broadcasting

- ITU BT.709, 2020, 2100 ...
- Over the air – traditional broadcast methods
 - legacy analogue – PAL, NTSC, etc
 - Cable – DOCSIS
- DVB-C, T, S, I etc...

Media Formats

- MPEG 2, 4, A, I
 - H.264, 265
- VP9 – AV-1

Transmission

- DVB
- SDI
- IP
- SDI on IP SMPTE 2022

Network

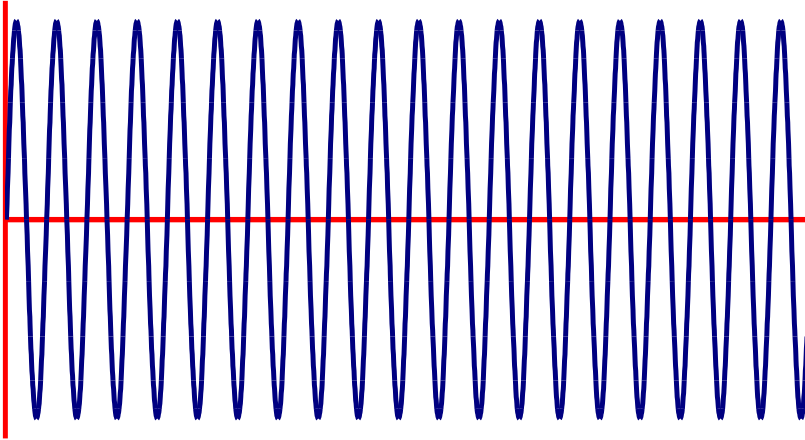
- ISO OSI 7 layer
 - MAC, Ethernet, IP, TCP, etc
 - ISDN, CCITT, etc
 - 3G, 4G, 5G ...
 - SDH, SONET

Workflow

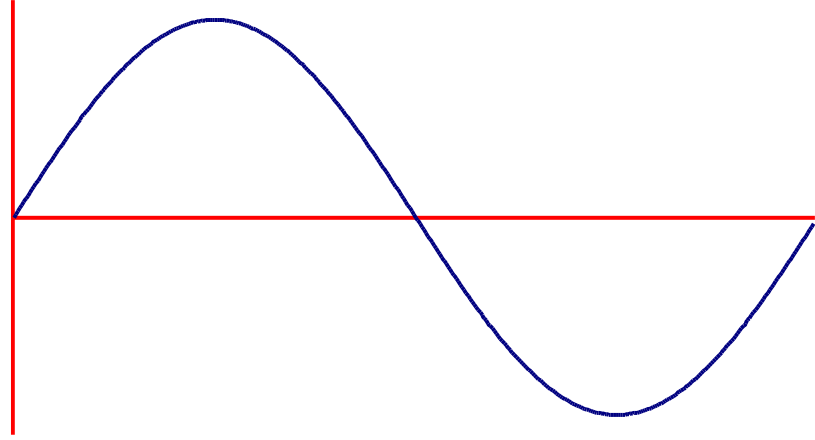
- MXF
- LTO
- SOA / IP
- Semantics – RDF, OWL
- Metadata – EBU Core, MPEG-I
- SMPTE 2022, 2110, 2059,
 - AES-67
 - AMWA NMOS

Amplitude Modulation

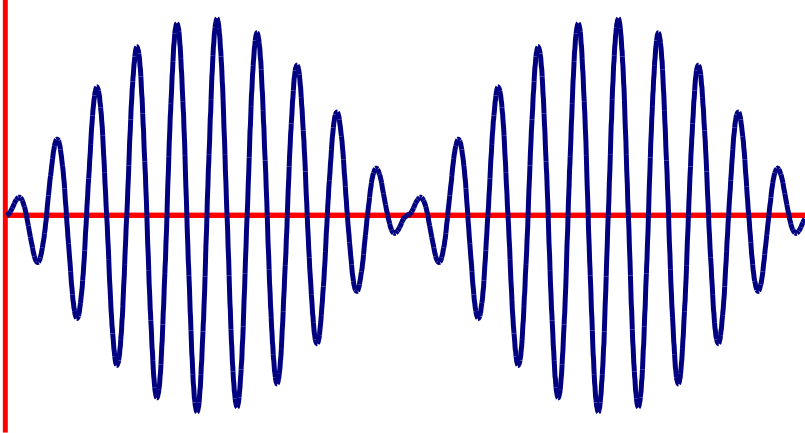
RF Carrier Wave



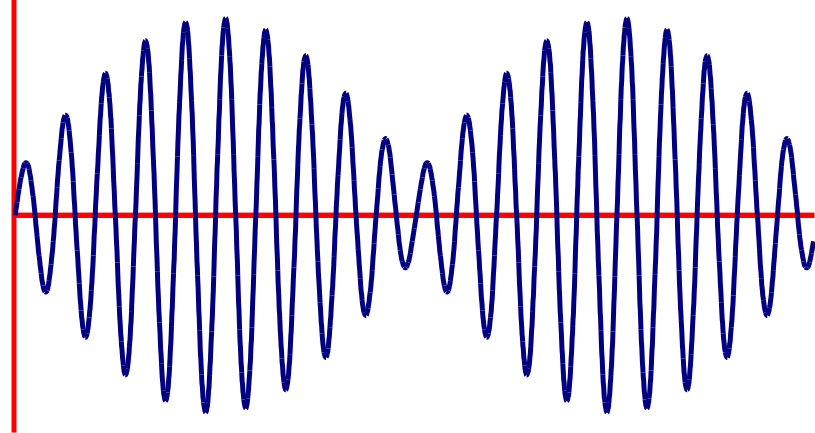
Modulation Information



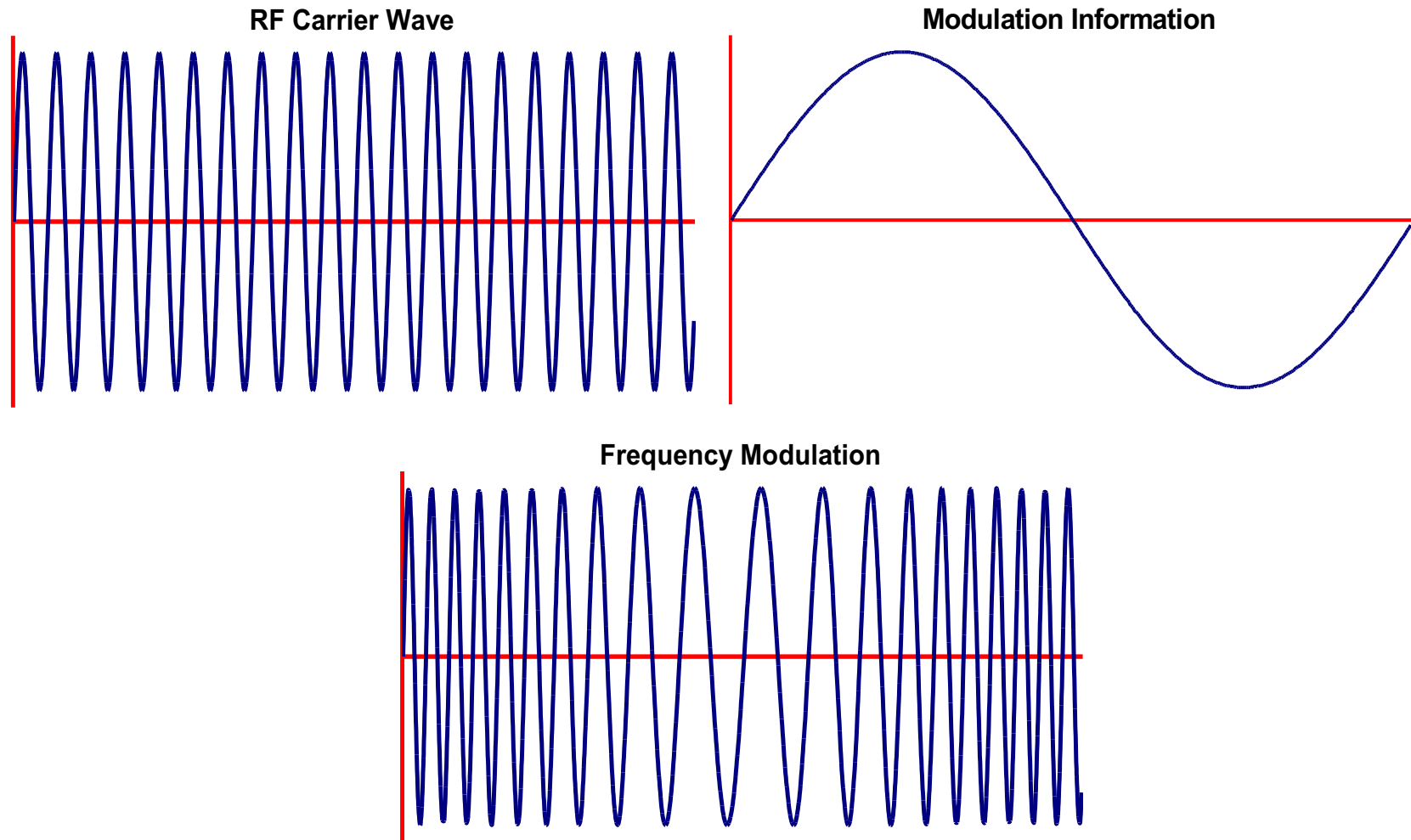
Amplitude Modulation



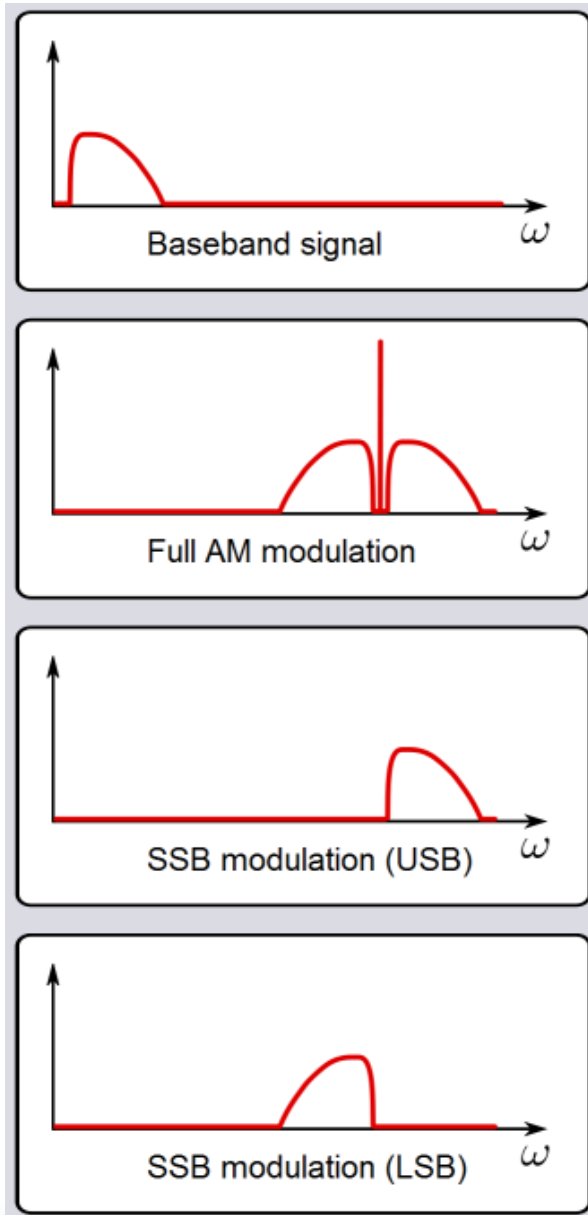
Amplitude Modulation (Min Carrier 20%)



Frequency Modulation

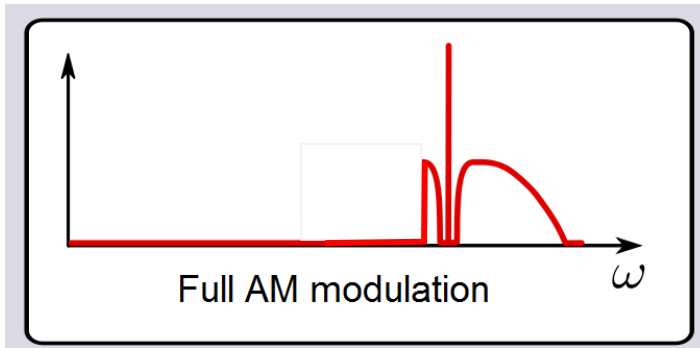


Sidebands – Single, Vestigial



- SSB produced by filtering one of the sidebands, usually the lower (LSB)
- Carrier is also often reduced or removed entirely (suppressed)
= single sideband suppressed carrier (SSBSC)
- No information is lost in the process
- Radio frequency transmission is now concentrated in a single sideband
 $\therefore$ effective power output is greater than in normal AM
(since the carrier and redundant sideband account for well over half of the power output of an AM transmitter)
- Requires special reception to demodulate

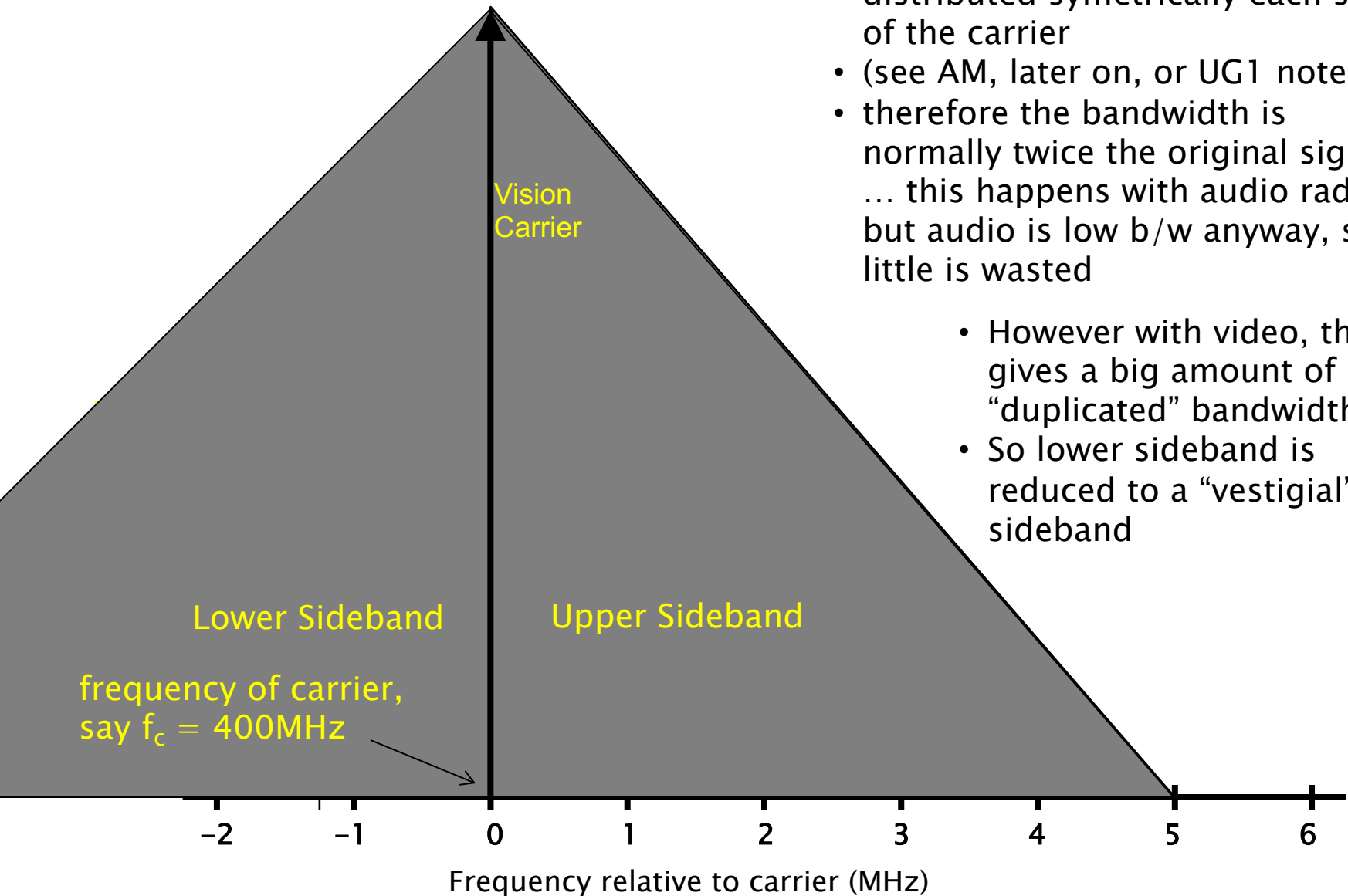
Vestigial Sideband (VSB)



- sideband that has been only partly cut off or suppressed
- to reduce the large bandwidth of the original video signal
- also be used in digital transmission, such as the ATSC standardized 8-VSB
- by the RF and IF filters

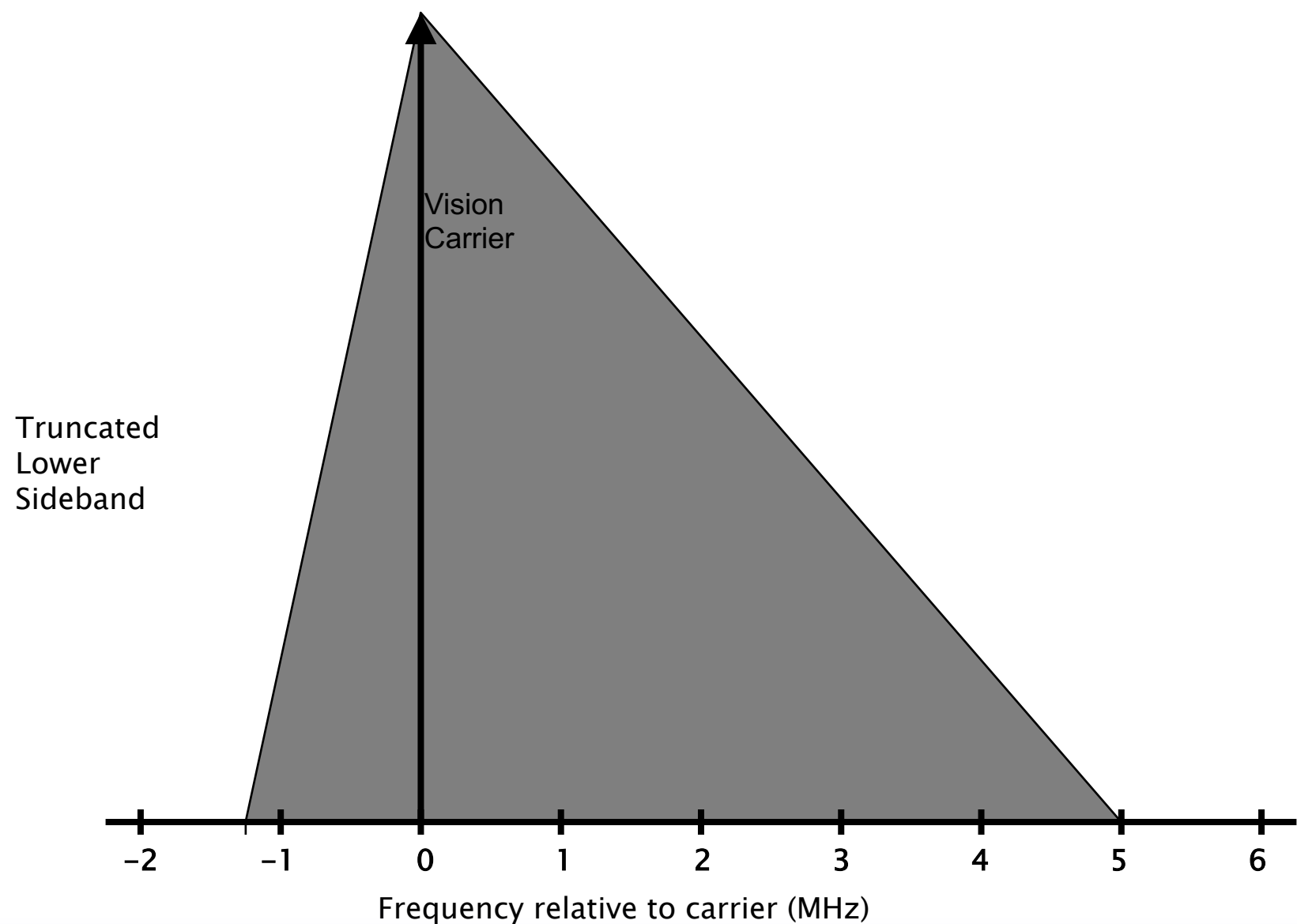
- For video – SSB would be desirable
... but the video signal has significant low frequency content, plus square sync pulses
... so some of the lower sideband is retained
- VSB – full upper sideband of bandwidth transmitted = 4MHz
... but only 1.25MHz of the lower sideband is transmitted, plus the carrier
- effectively makes the system AM at low modulation frequencies and SSB at high modulation frequencies
- absence of the lower sideband components at high frequencies must be compensated for, and this is done by the RF and IF filters

PAL-B Spectrum

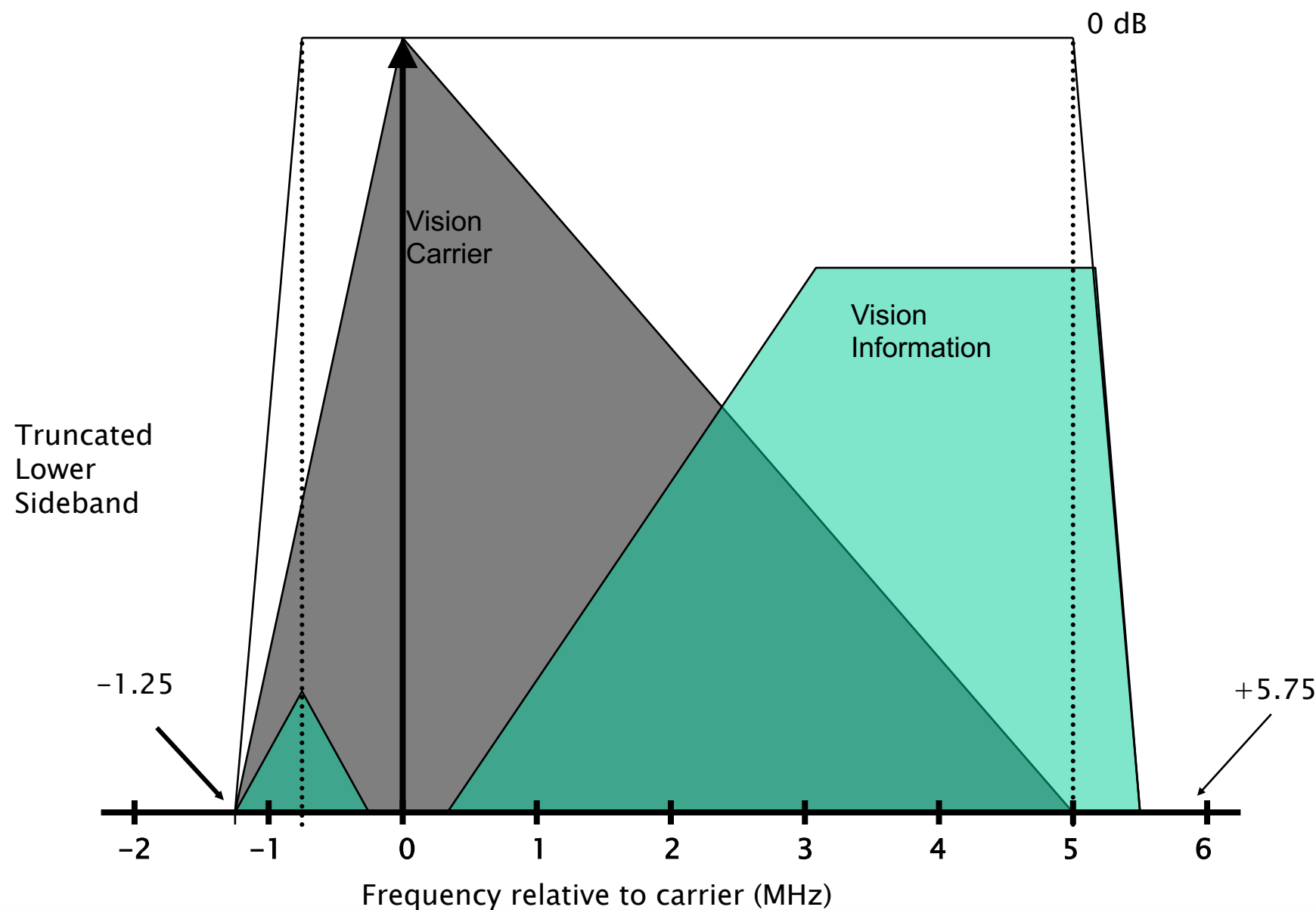


- AM information is normally distributed symmetrically each side of the carrier
- (see AM, later on, or UG1 notes)
- therefore the bandwidth is normally twice the original signal ... this happens with audio radio, but audio is low b/w anyway, so little is wasted
- However with video, this gives a big amount of “duplicated” bandwidth
- So lower sideband is reduced to a “vestigial” sideband

PAL-B Spectrum

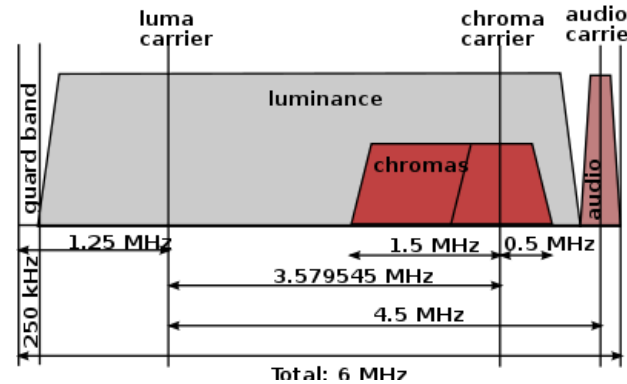


PAL-B Spectrum

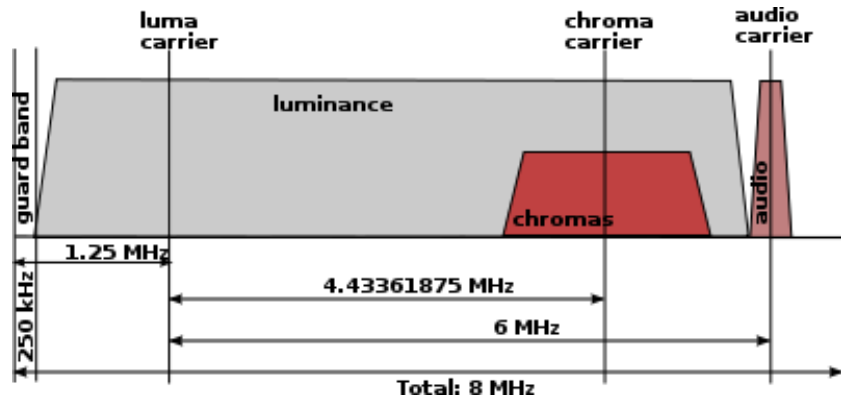


PAL, NTSC and SECAM Spectra

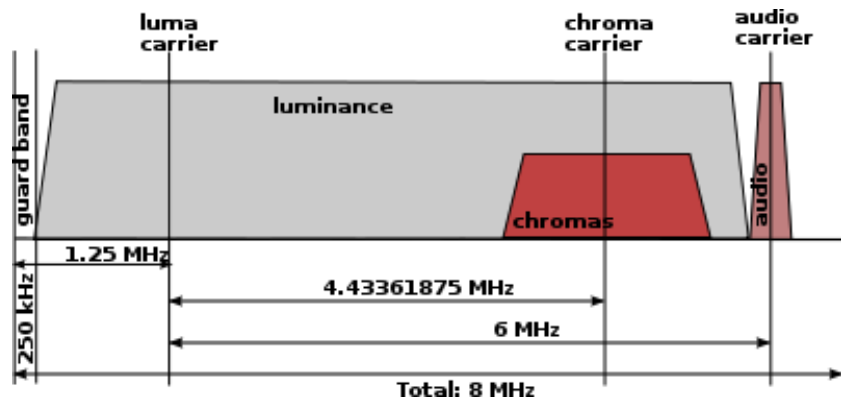
NTSC



PAL

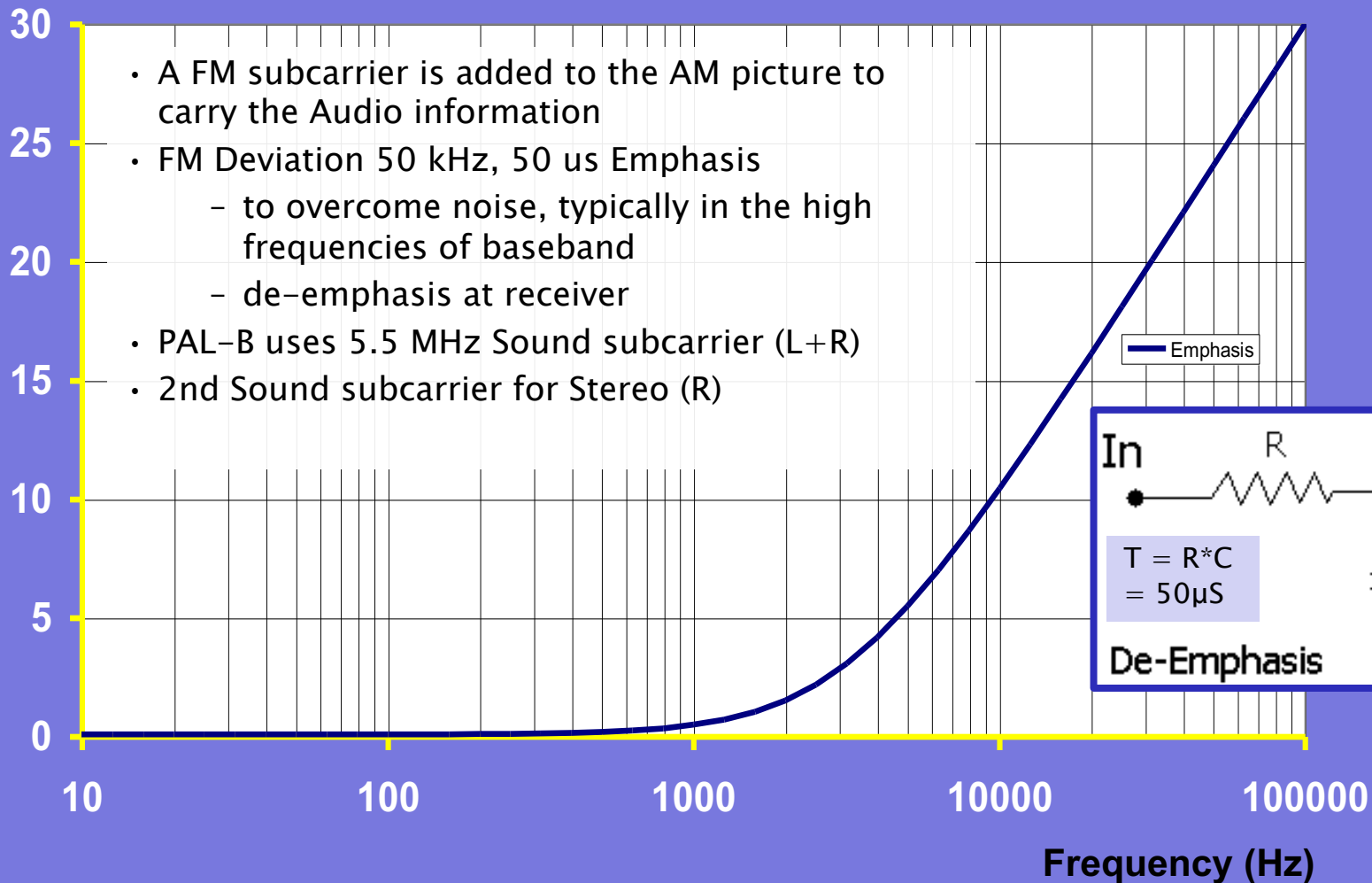


SECAM

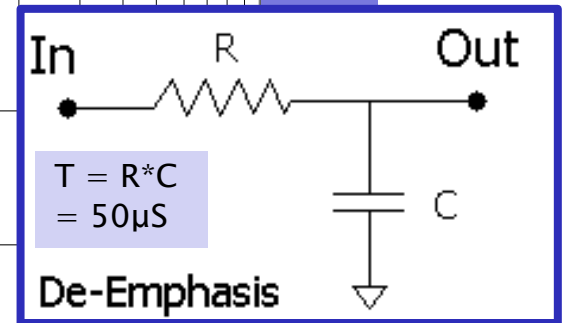


FM Sound Emphasis

50 us Emphasis

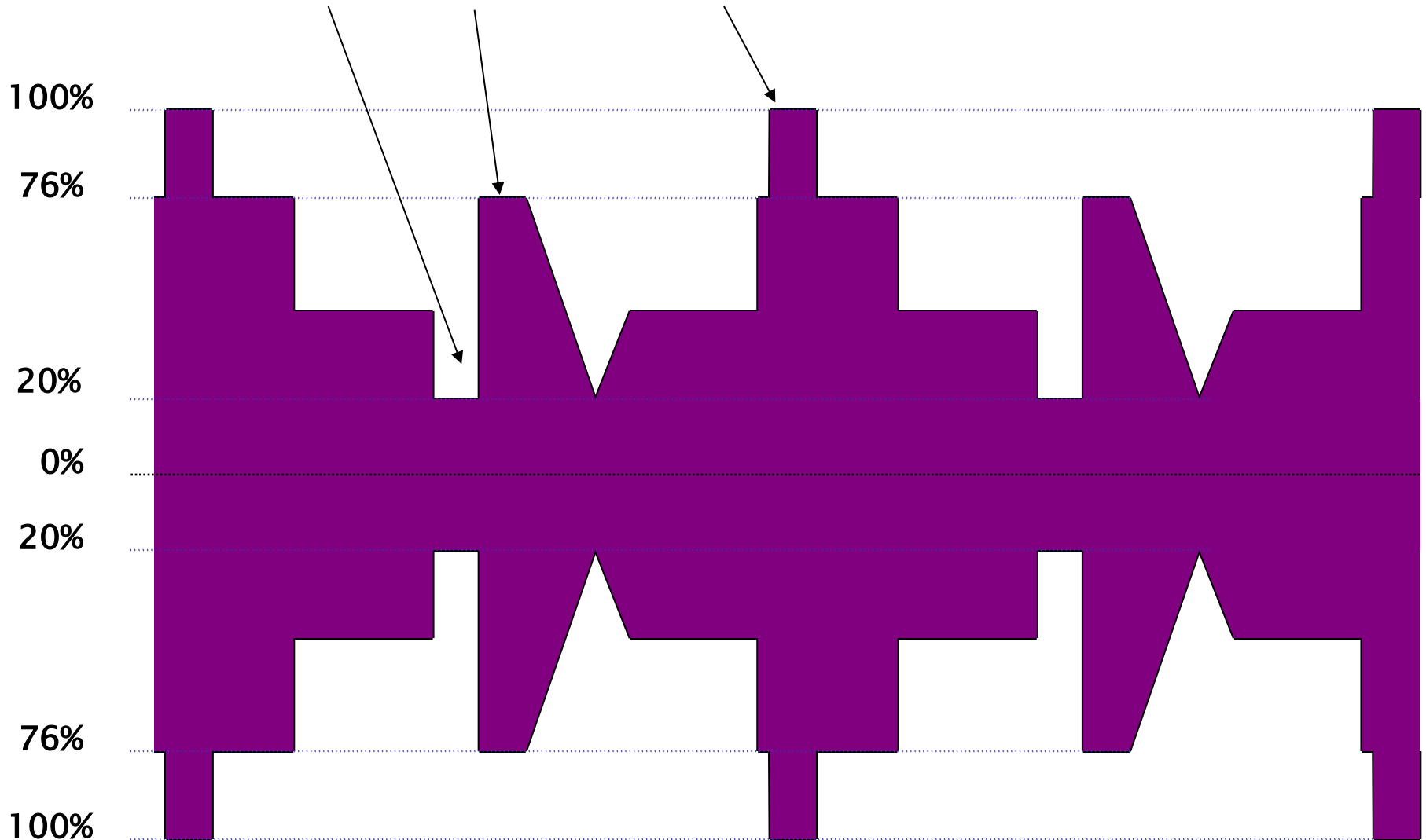


- A FM subcarrier is added to the AM picture to carry the Audio information
- FM Deviation 50 kHz, 50 us Emphasis
 - to overcome noise, typically in the high frequencies of baseband
 - de-emphasis at receiver
- PAL-B uses 5.5 MHz Sound subcarrier (L+R)
- 2nd Sound subcarrier for Stereo (R)



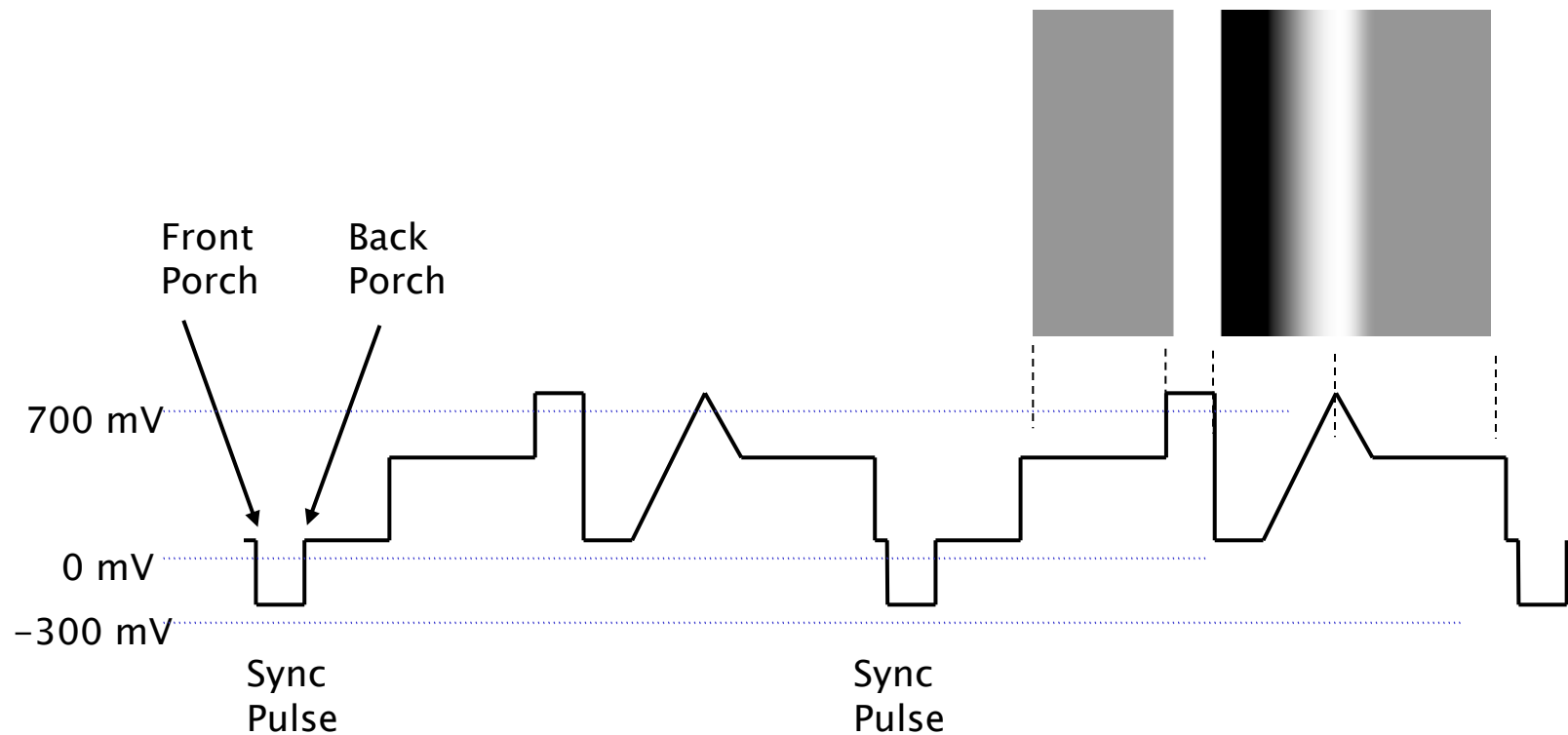
TV Modulation – AM; Min 20%

- Peak White 20% Black 76% Syncs 100% *Note – “Inverse modulation”*



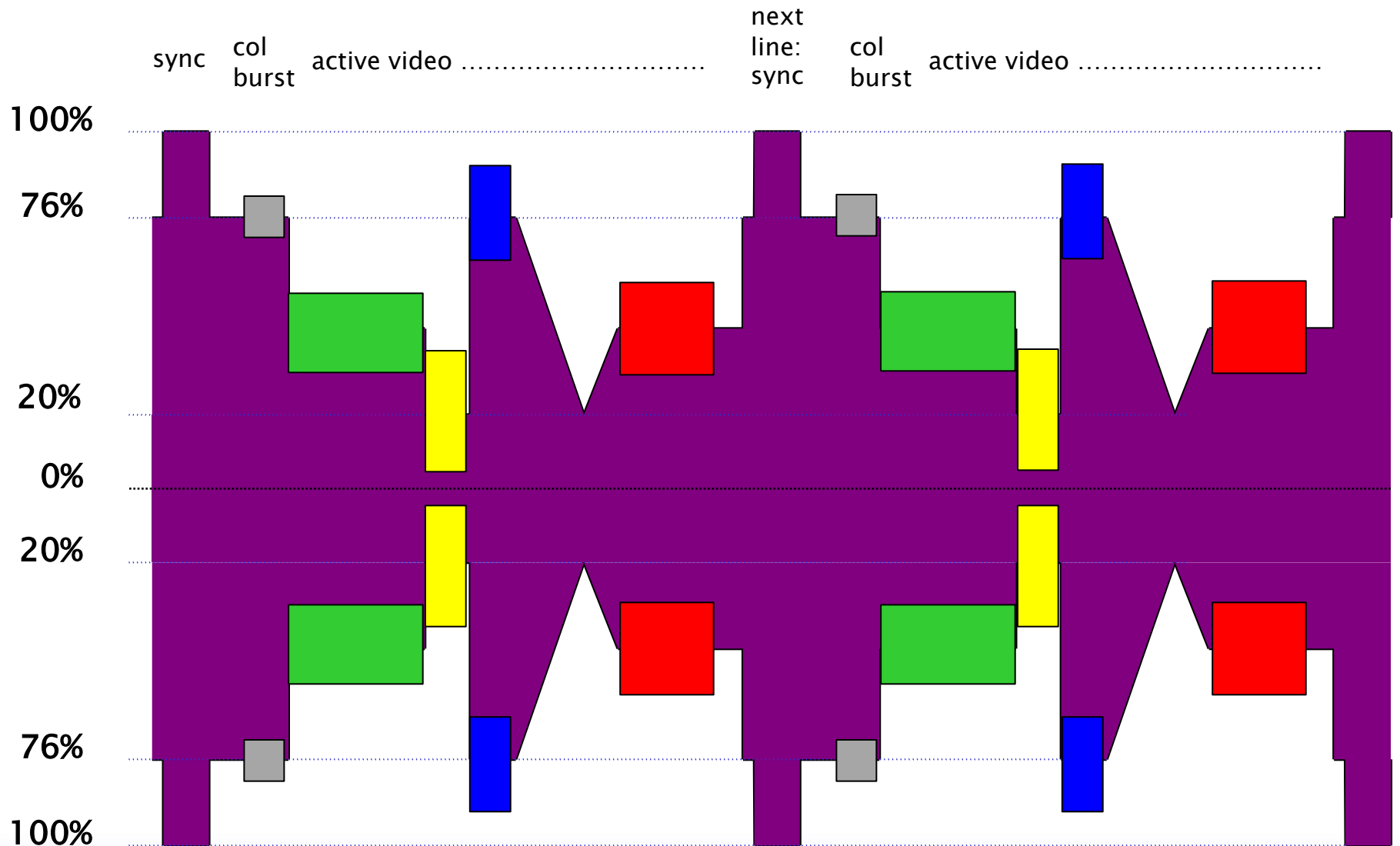
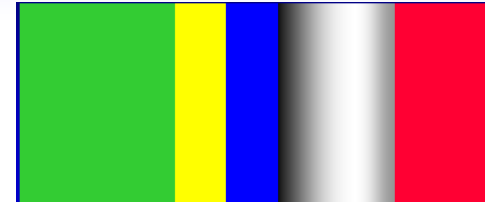
The Analogue Video Signal

- First Television Pictures were Black & White
- Referred to as Luminance
- Video refers to the linear base-band signal that contains the image information

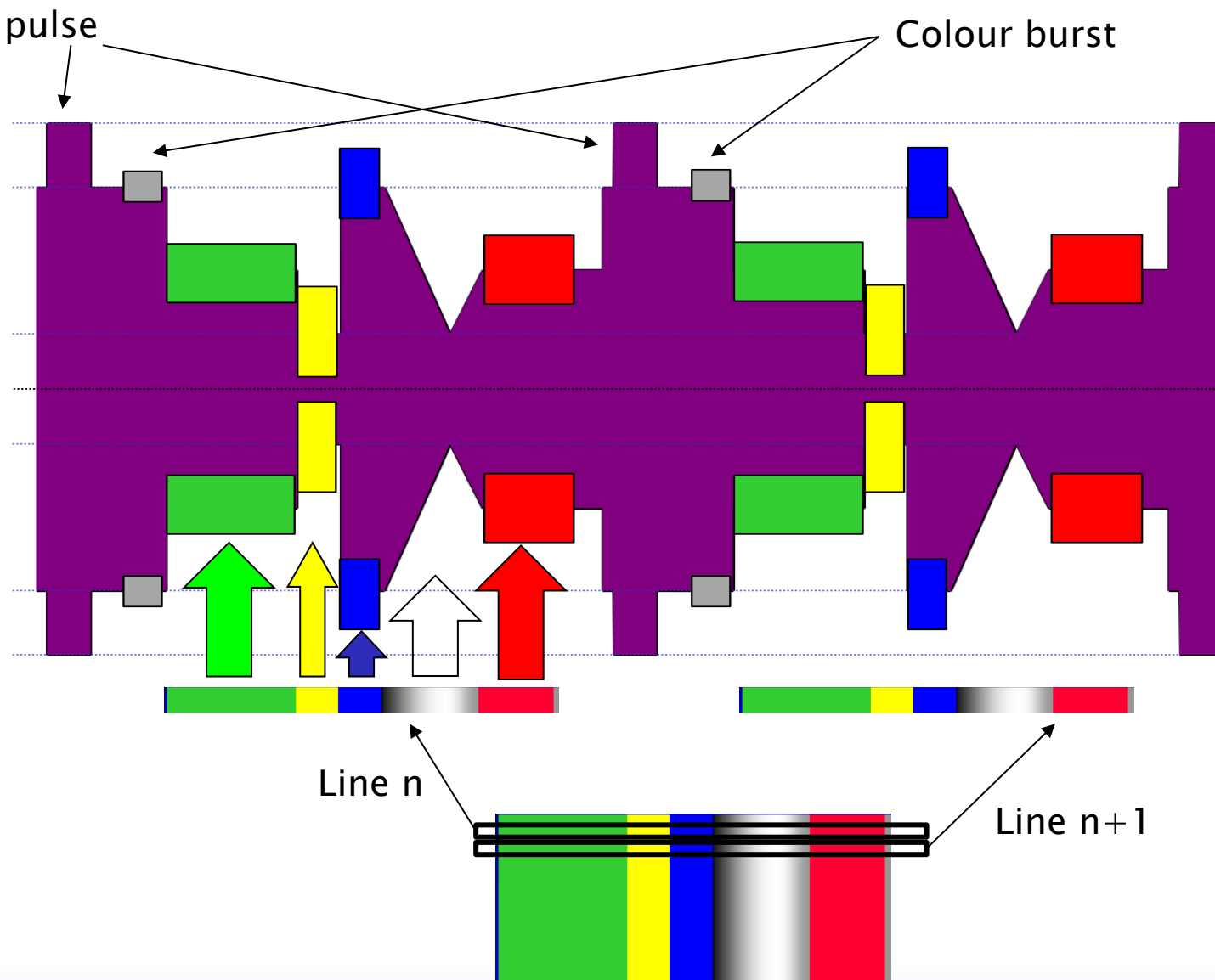


TV Modulation – PAL, AM

Colour signal added to AM signal



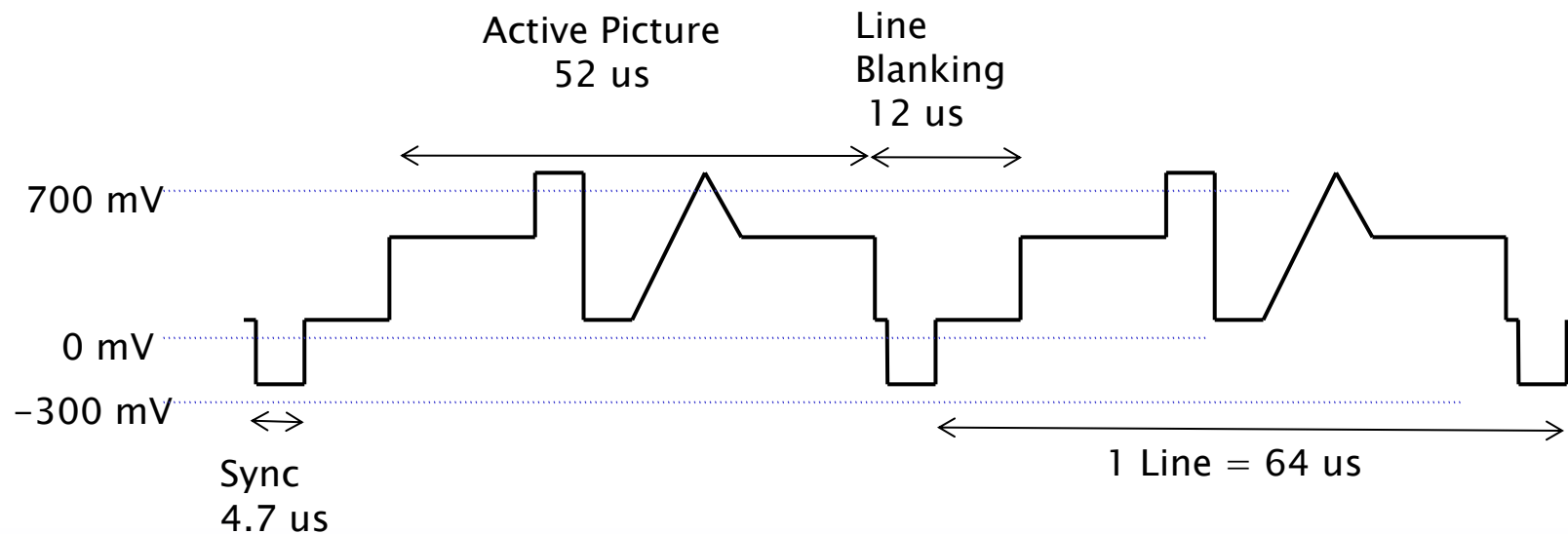
TV Modulation – PAL, AM



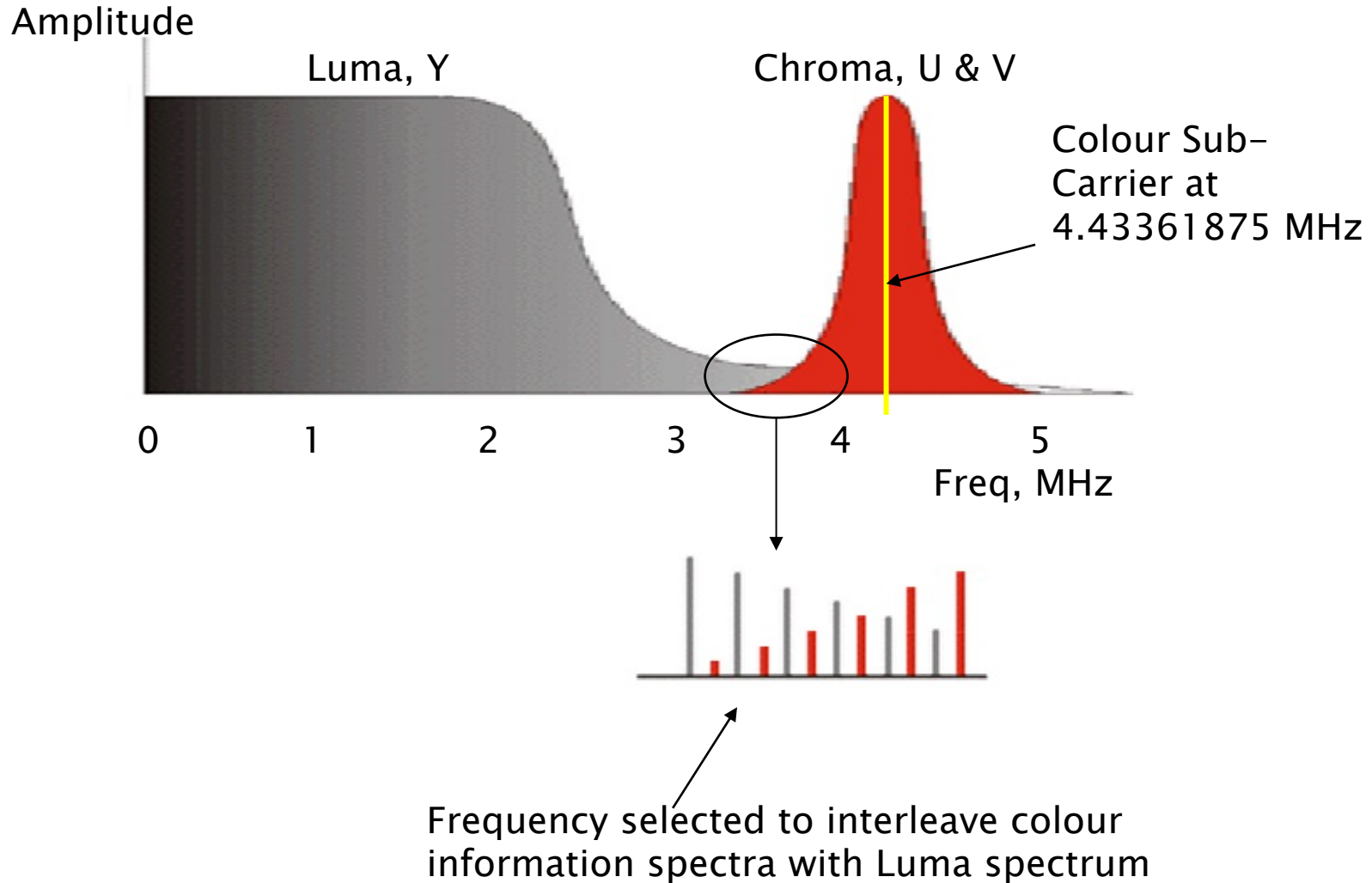
Video Timing

SDTV

- 64 μ s for each line (15.625 kHz)
- 52 μ s Active Picture Area
- 12 μ s Blanking and Synchronisation
- Two level sync pulse 300 mV below blanking



Colour Sub Carrier

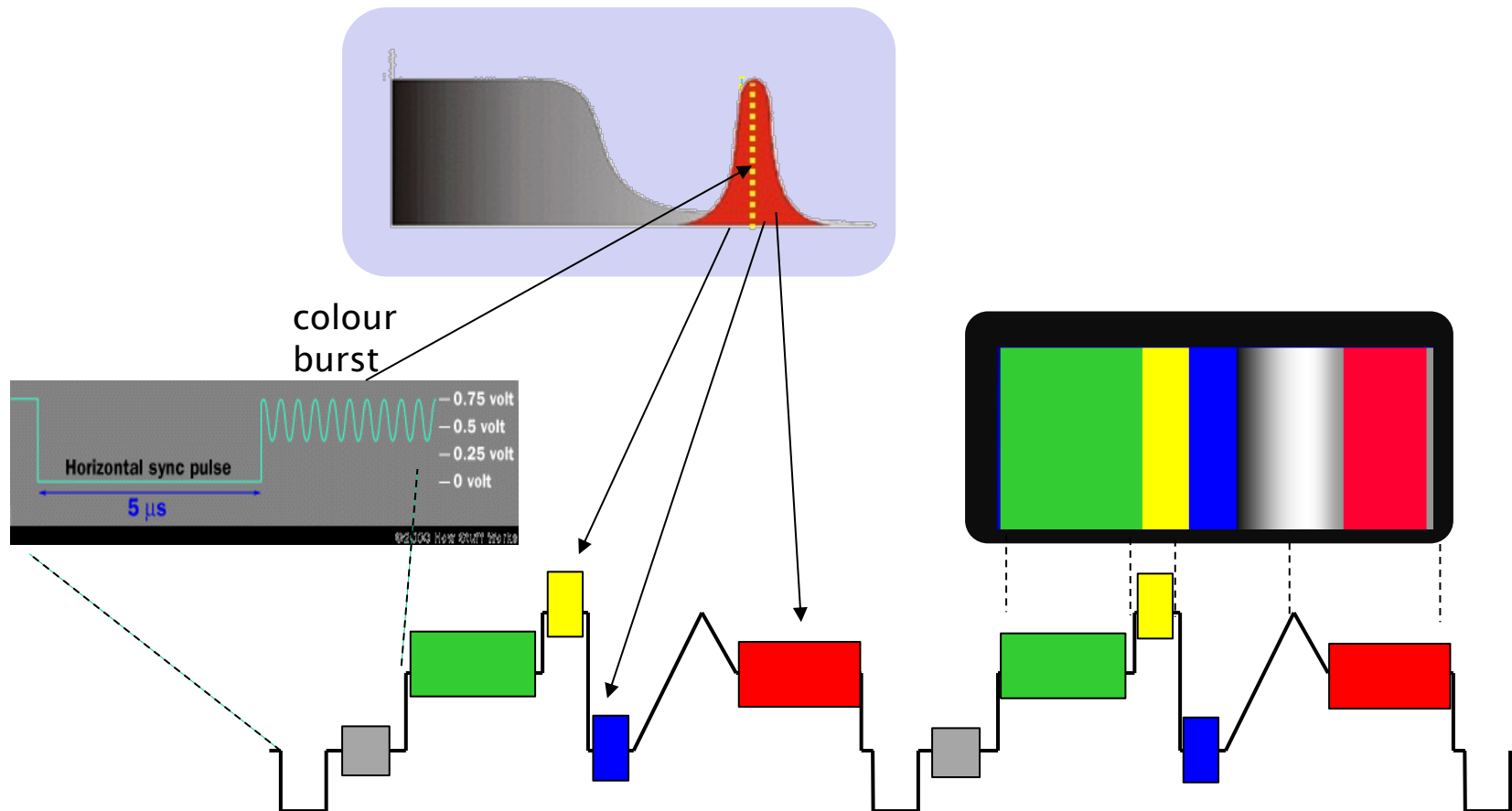


Adding Colour to B&W Video

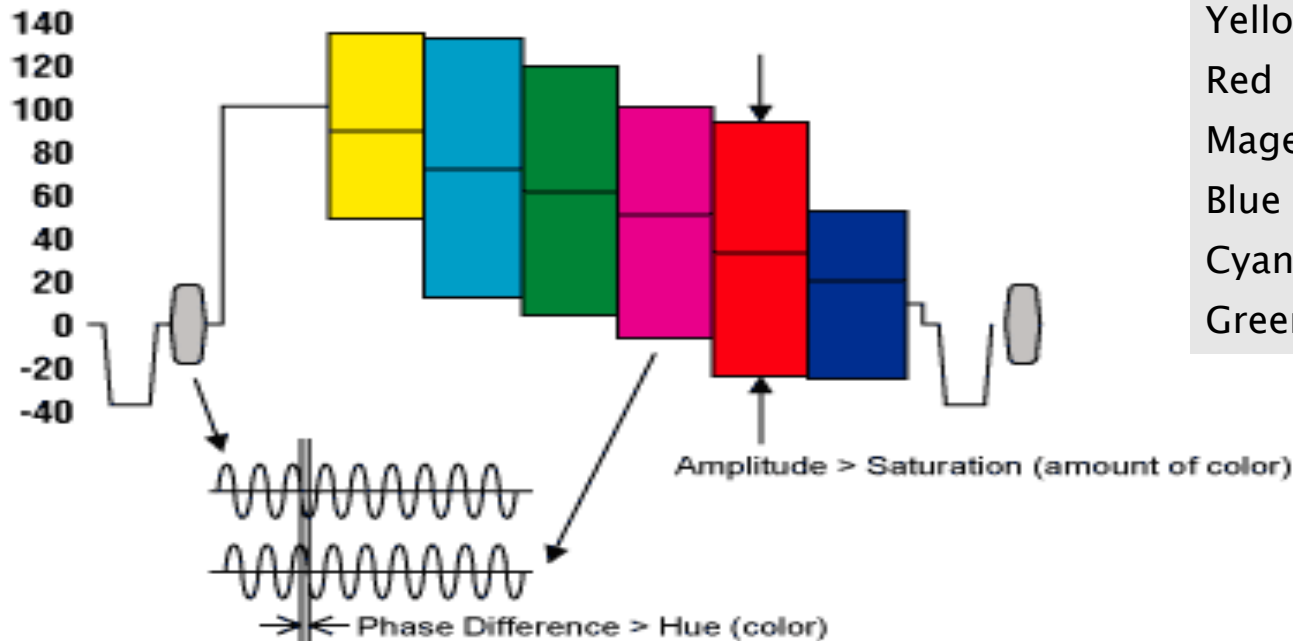
Following Sync pulse, eight cycles of a sine wave are added as a colour burst to sync colour carrier (suppressed) to decode colour info. Carriers:

NTSC = 3.579545 MHz

PAL = 4.433618.75 MHz



PAL Colour Encoding



Color	Phase
Burst	0 degrees
Yellow	15 degrees
Red	75 degrees
Magenta	135 degrees
Blue	195 degrees
Cyan	255 degrees
Green	315 degrees

The hue of a given point on the screen

- defined by the phase of the subcarrier at that point
- with reference to the phase of the carrier insertion oscillator
- which is synchronised to a burst of subcarrier at the start of each line

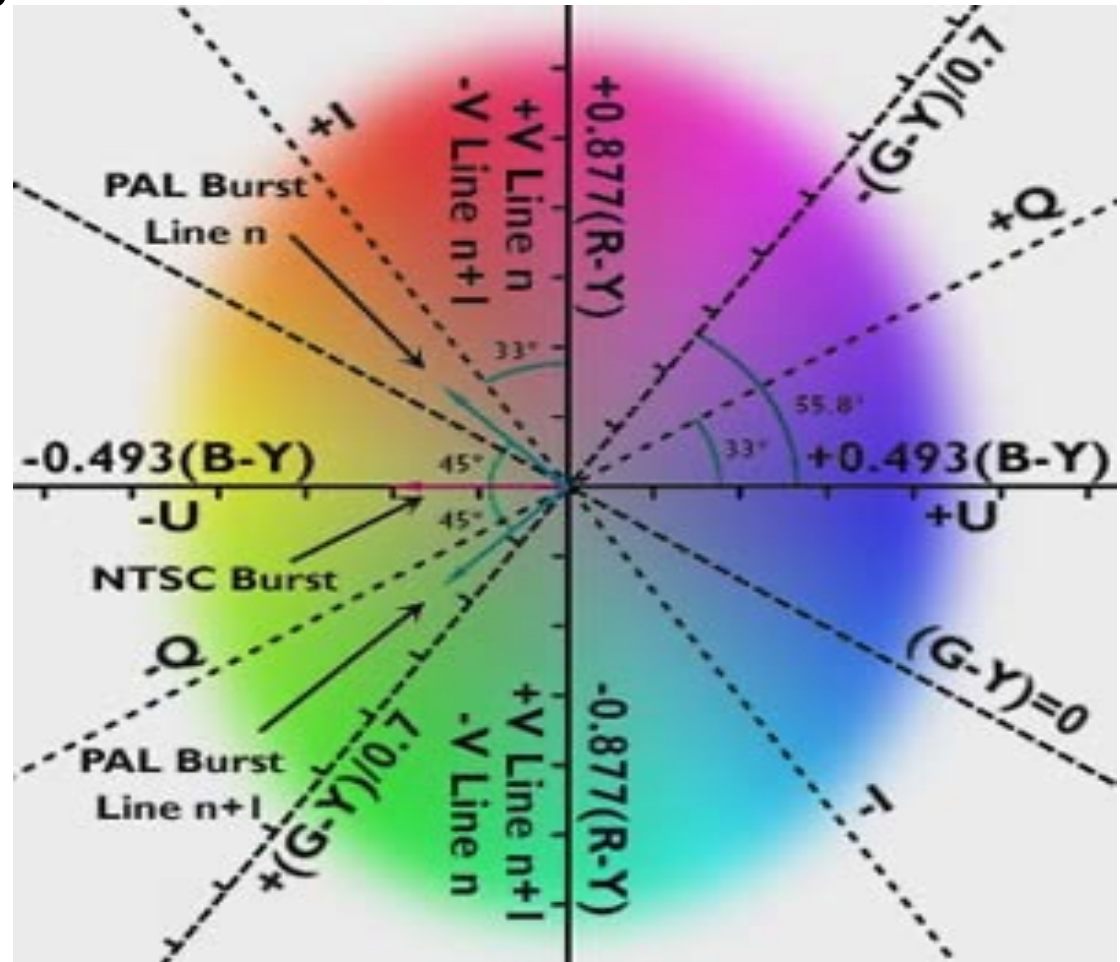
Hue errors

when the phase of the colour subcarrier has shifted during transmission with respect to the phase of the reference burst.

<https://www.maximintegrated.com/en/app-notes/index.mvp/id/734>

PAL Colour Encoding

Phase shift of the carrier in accordance with the PAL colour bar signal measured vector standard



PAL vs NTSC Colour

NTSC

- colour instabilities due to multipath reception and transmitter imperfections
- receivers needed a hue control to compensate

PAL early 1960's – developed to avoid NTSC issues

- increased bandwidth – better picture quality
- to correct shifts in the colour sub-carrier phase of the signal
- every other line has its phase in reverse order = Phase Alternate Line
- aimed to keep out non-European manufacturers from the TV set market
- Japanese managed to design decoders which circumvented the PAL patents by treating the signal like NTSC but decoding only every other line

France – SECAM – sequential couleur avec memoire (sequential color with memory)

- same bandwidth as PAL; transmits the colour information sequentially.
- Eastern Block adopted for purposeful incompatibility with western transmissions (politics)

NTSC – why 29.97 fps?

.. and how did PAL get it right?

Matt Parker explains ...

<https://www.youtube.com/watch?v=3GJUM6pCpew>



Standardising Colour
Standardising TV formats
Recs BT.709, BT.2020, BT.2100

B B C HD 1080 lines

RGB Colour Space

variety of standards for mapping colour

most popular –

CIE 1931 – widely used

CIE 1976 – strongly advised

CIE – Commission internationale de l'éclairage

Gamut – the achievable colours in the colour space;

Chromaticity diagrams

- measure colour quality, independent of other factors like luminance
- colour space is a uniform representation of visible light
- maps the all of the colours visible to the human eye onto an x–y grid
- assigns them measureable values
- allows uniform measurements and comparisons between colours
- offers certainty that images look the same from display to display when used to create colour gamut standards

CIE Chromaticity Diagram



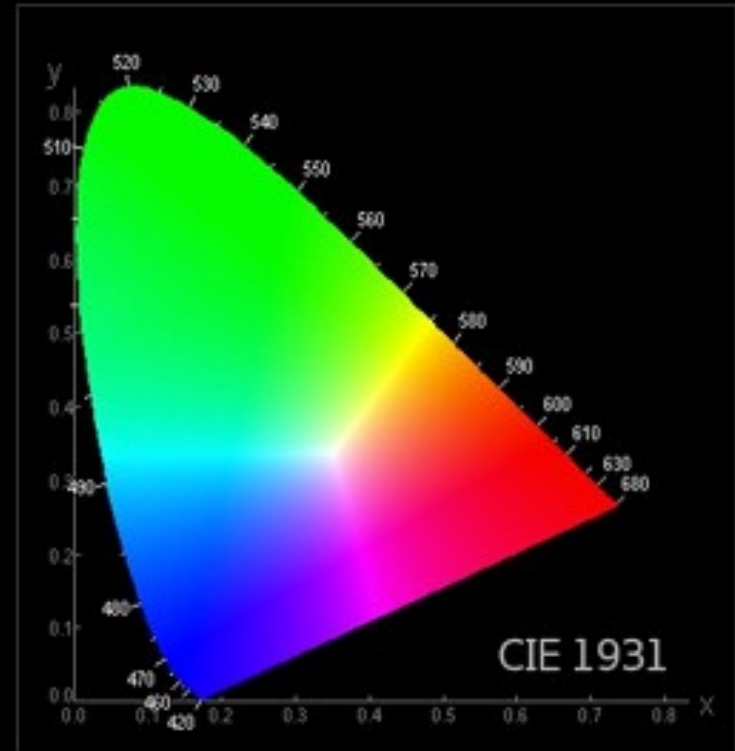
Chromaticity and CIE Colour Space

Rec. 709 – for monitors, not cameras

- defines a gamma of 2.2

CIE spaces

- Colours defined by their coordinates on a CIE chromaticity diagram
- diagram deals with hue and saturation, not brightness; colour defined by two numbers
 - R: $x_R = 0.64$ $y_R = 0.33$
 - G: $x_G = 0.30$ $y_G = 0.60$
 - B: $x_B = 0.15$ $y_B = 0.06$
- also defines
 - white: $x_W = 0.3127$ $y_W = 0.3290$
- establishes the effective colour temperature of the display
- e.g. rec 709 cannot display a green deeper than 0.3, 0.6 on CIE 1931



BT.709

HDTV systems 2M luma samples / frame.

Rec. 709 two parts:

- 1080i and 1080p systems
- 1035i30 and 1152i25 HDTV systems
- 1035i30 now obsolete; 1152i25 experimental, never commercially deployed

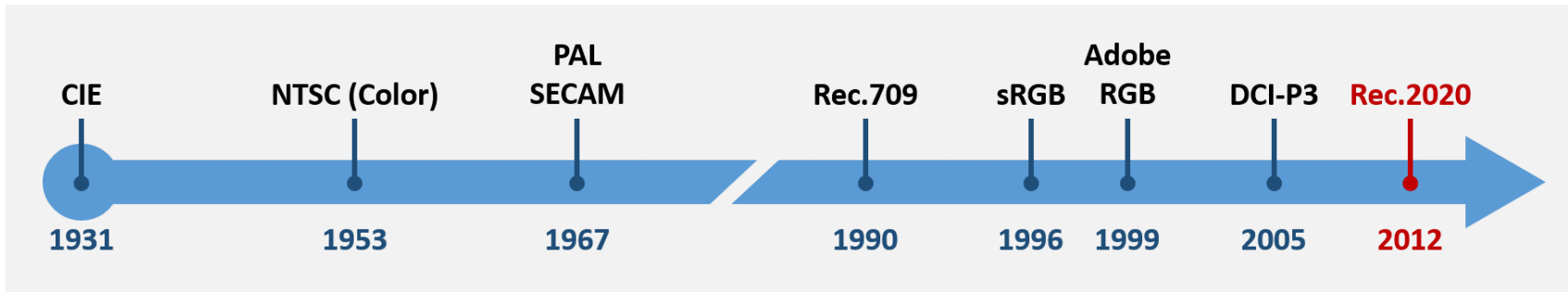
Frame Rate

- picture rates: 60 Hz, 50 Hz, 30 Hz, 25 Hz and 24 Hz.
- "Fractional" rates (divided by 1.001) also permitted

Digital Representation

- R'G'B' and Y'CBCR
- 8 or 10 bits per sample in each color channel
- 8-bit encoding: R', B', G', and Y' channels range of 16 ... 235
 - Cb, Cr channels = 16 ... 240 (128 = neutral value)
 - R'G'B' – reference black = 16, 16, 16; reference white = 235, 235, 235
 - Y'CbCr – reference black = 16, 128, 128; reference white = 235, 128, 128
 - Values outside nominal ranges allowed, but usually clamped for broadcast
 - 0 and 255 = timing references
- For 10-bit encoding – 4 x (8-bit encoding values)
... [i.e. 2 extra bits = 4x]

Colour Standards



NTSC / PAL / SECAM ... *CIE Coverage: 47%*

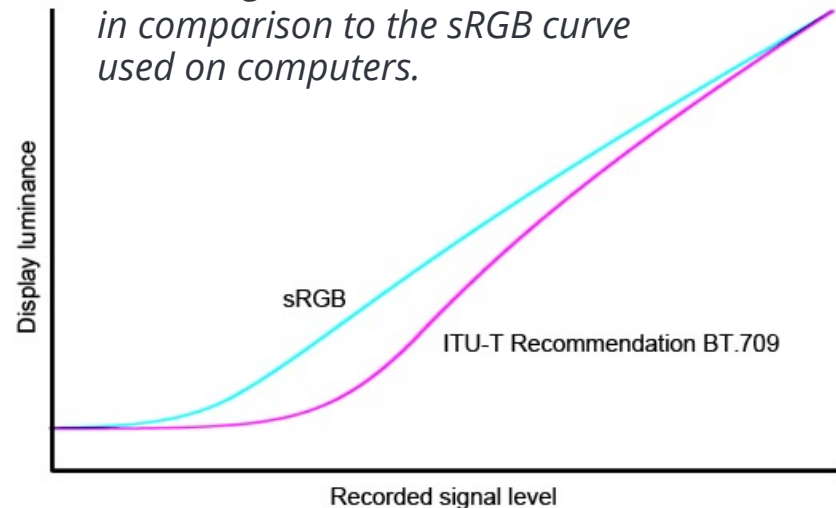
- primary standards since 1950s for analogue broadcasting
- largely phased out due to digital HD broadcasting.

<http://www.redsharknews.com/technology/item/3289-what-exactly-is-rec-709>

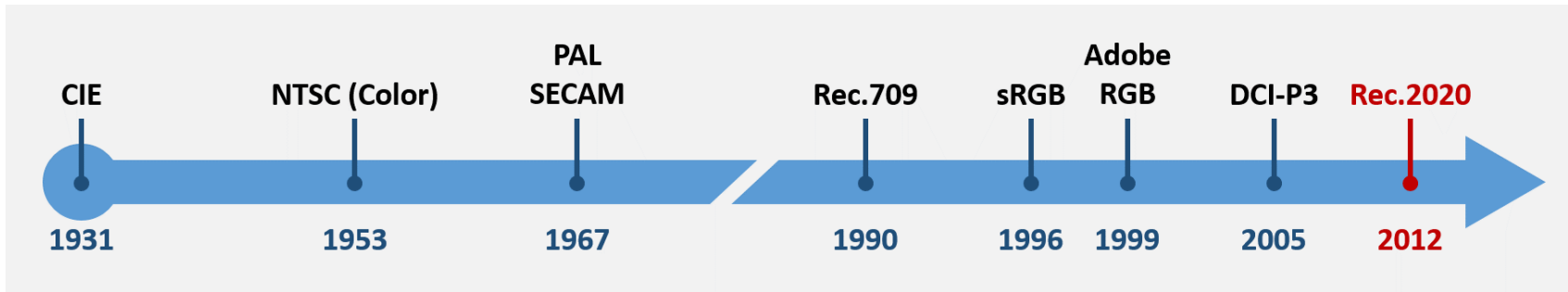
Rec BT.709 ... *CIE Coverage: 34%*

- 1990, ITU; predominate standard for digital HD broadcasting and TVs (also DVD and Blu-ray)
- standard for HD format, aspect ratio, 709 colour space
 - actually covers less of the CIE than the NTSC

Rec. 709 gamma curve, shown here in comparison to the sRGB curve used on computers.



Colour Standards



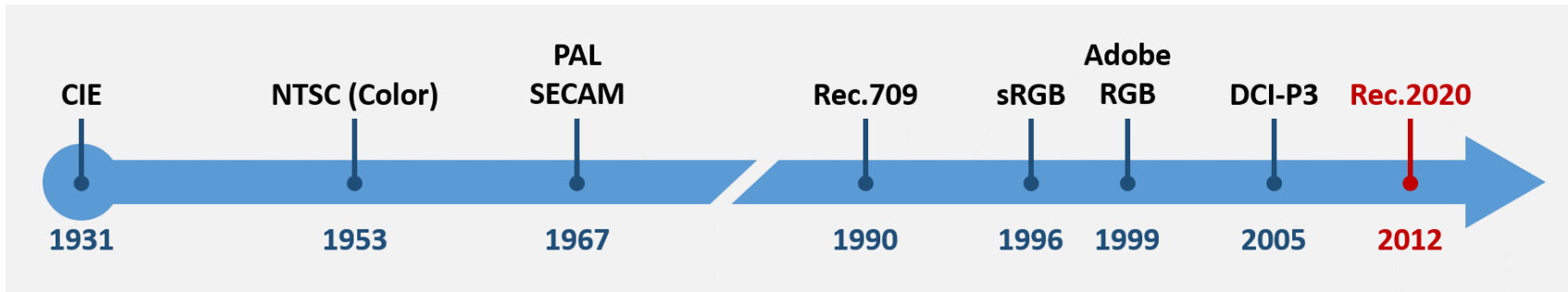
sRGB ... CIE Coverage: 34%

- 1996, HP and Microsoft
- colour gamut identical to Rec.709
- most commonly used standard worldwide for just about every device other than TVs
- for internet streaming, monitors, projectors, cameras, printers, game consoles, smart phones, tablets, touch screen, etc

Adobe RGB ... CIE Coverage: 50%

- 1999, Adobe
- to address accurately matching display colour with CMYK printing color
- now increasingly preferred in creative industries due wider gamut than sRGB.

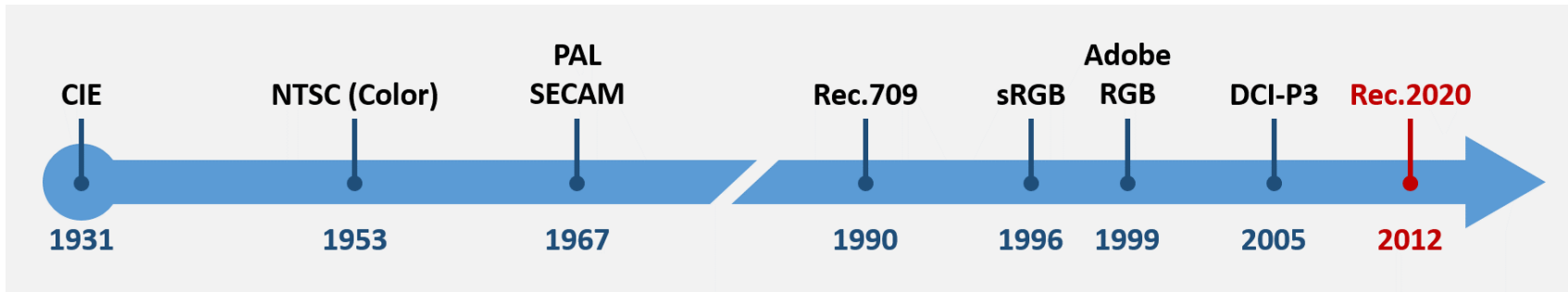
Colour Standards



DCI-P3 ... CIE Coverage: 46%

- 2005, consortium of cinema, television and entertainment companies
- to ensure consistent content quality from original capture to final projection.
- preferred in creative industry due to emphasis on digital production and distribution
- used in cinema-grade cameras, projectors and projection equipment

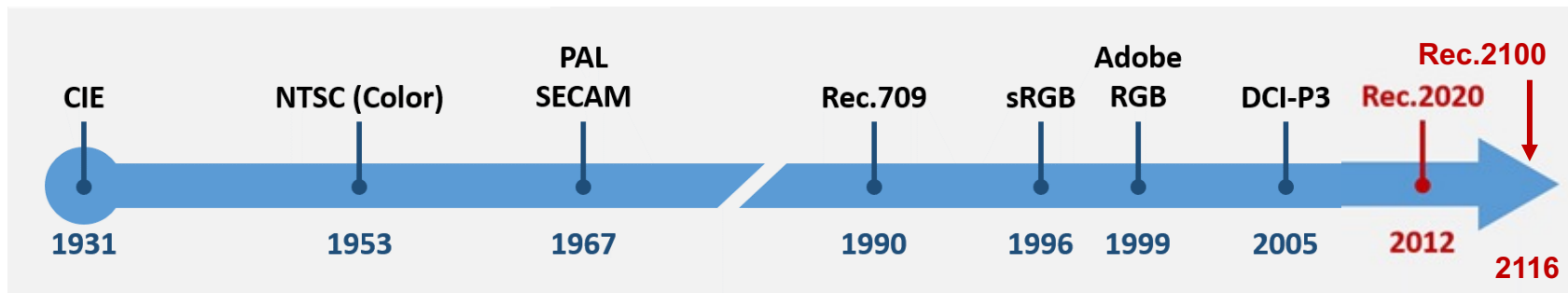
Colour Standards



Rec. BT 2020 ... CIE Coverage: 67%

- 2012, ITU; optimized for UHD. Covers many aspects, including wide colour gamut
- 68 *billion* colours.
- NHK – 2020 Olympics in full Rec.2020 colour
- Blu-Ray Disc Association – Rec.2020 for Ultra HD Blu-Ray
- Consumer Electronics Association (CEA) – definition of HDR – reference to Rec. 2020 primaries in its HDR10 Media Profile
- *resolution, frame rates, etc – BT.2100 expands on 2020 .. see over ...*

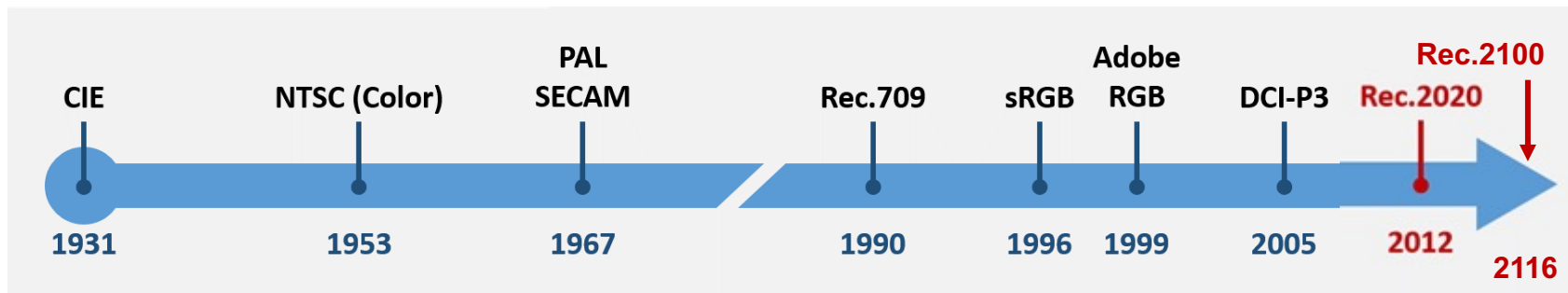
Colour Standards



BT.2100 – expands on BT.2020 (resolutions and colour gamut)

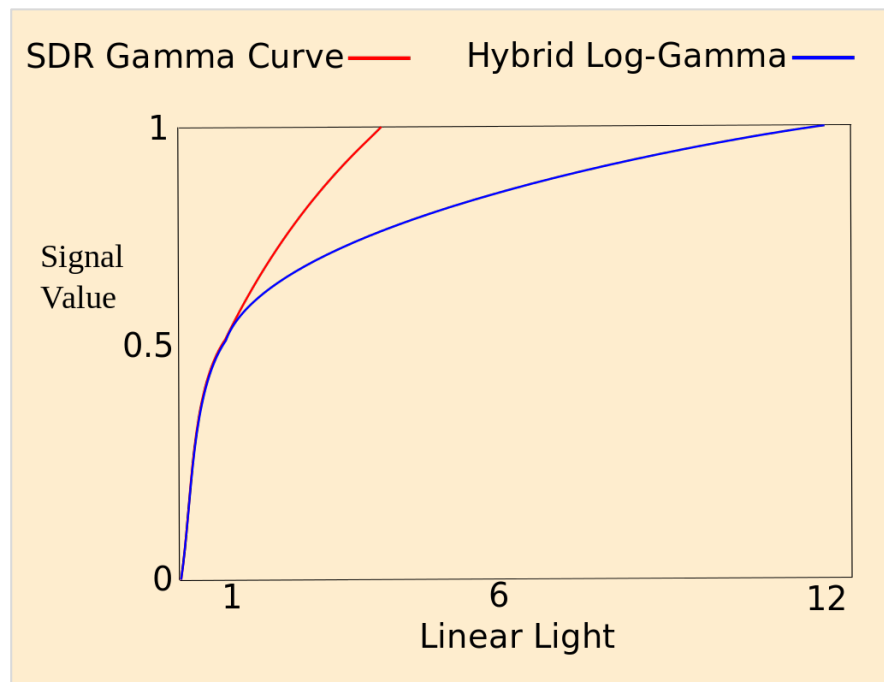
- 2016, ITU
- resolution: 1080p (x 1920); 3840×2160 (i.e. 4K); 7680×4320 (i.e. 8K)
 - square pixels
- frame rates: 120p, 119.88p, 100p, 60p, 59.94p, 50p, 30p, 29.97p, 25p, 24p, 23.976p
 - *only progressive scan frame rates*
- bit depth: 10 or 12
- narrow range colour or full range colour:
 - 10 bit (i.e. 0 – 1023), full range – all levels 0–1023
- same colour space as BT.2020

Colour Standards



Two transfer function formats

- **Perceptual Quantisation (PQ)**
 - more prevalent in the consumer equipment
 - HDR10 and Dolby Vision
- **Hybrid Log-Gamma (HLG)**
 - BBC, NHK
 - compatibility with legacy SDR displays
 - closely matches established television gamma systems

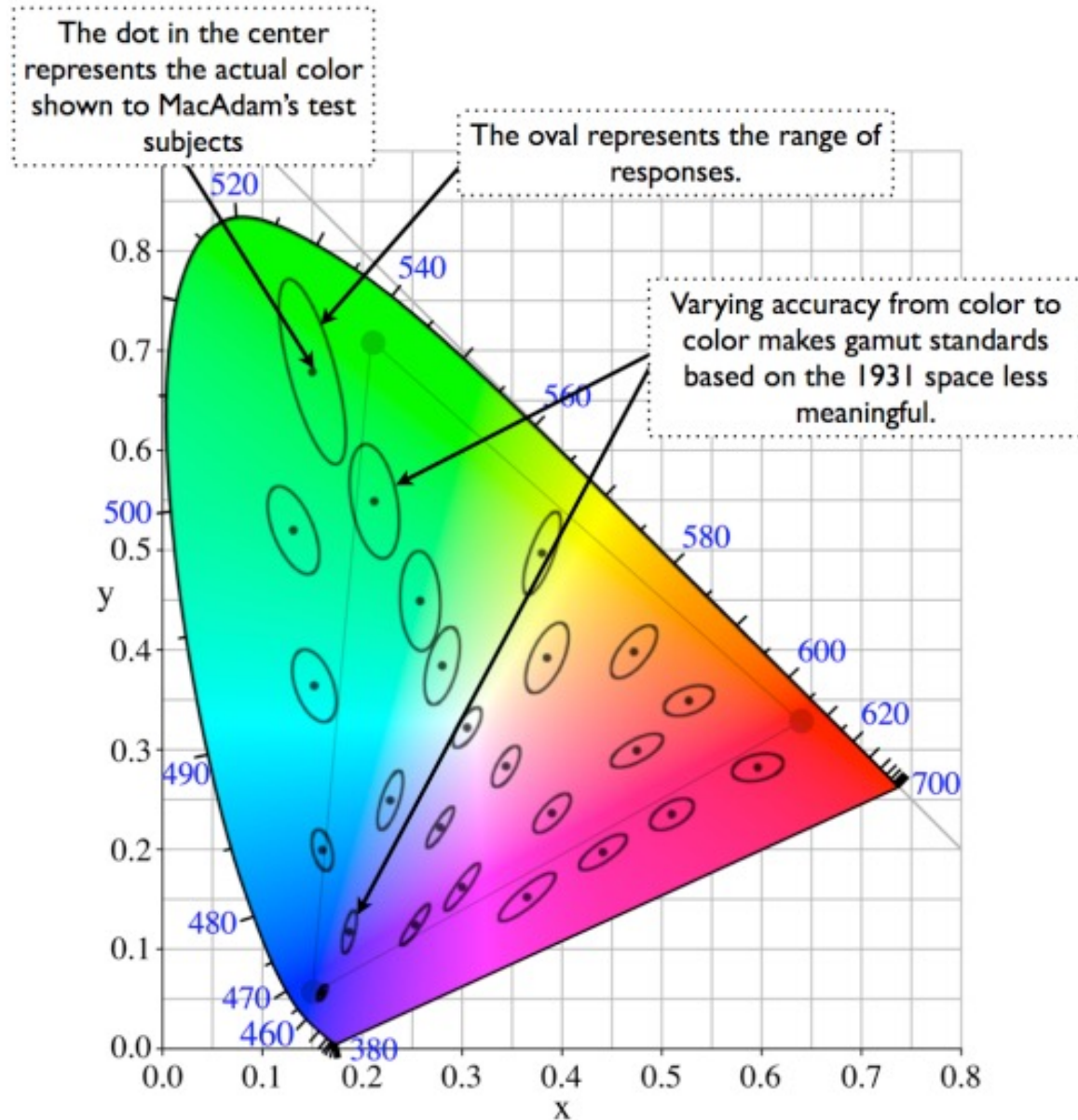


CIE1931 vs CIE1976

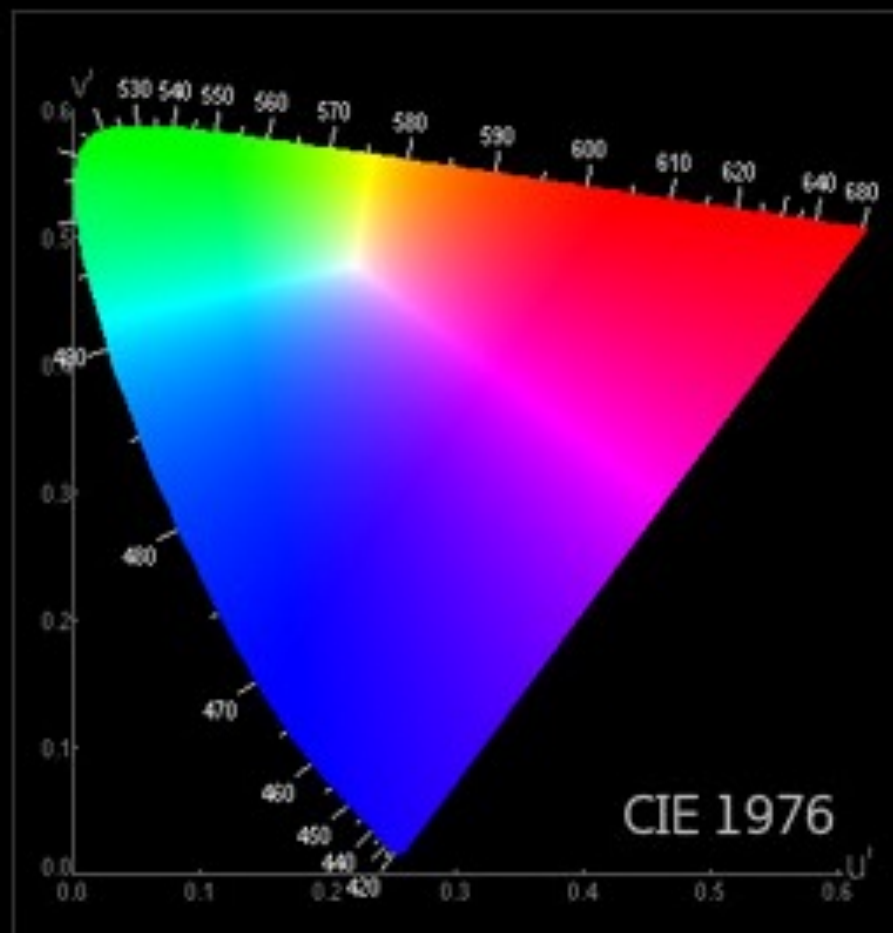
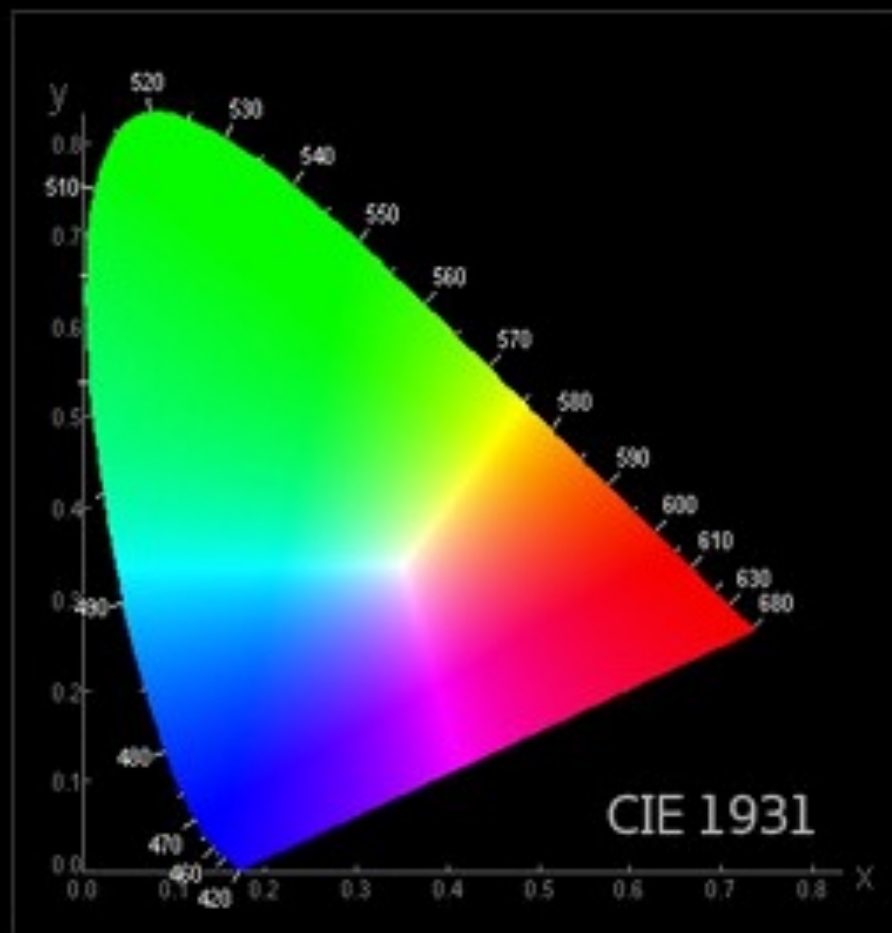
MacAdam (in mid 1970's)
“ ... CIE 1931 not linear,
e.g.
green – wide perceived
range of hues of green
blue – much more
agreement

MacAdam Ellipses
ellipse = the range of
colours respondents
reported

... so CIE1976
more linear colour space



CIE1931 vs CIE1976



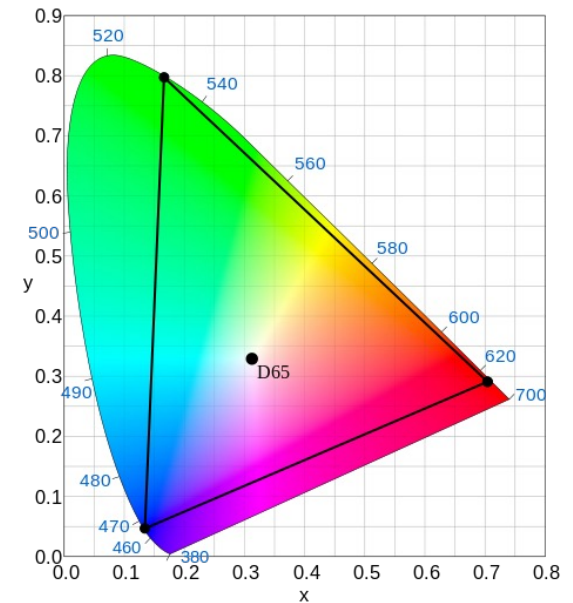
ITU Rec 2020

ITU-R Rec BT.2020

August 2012

defines aspects of UHDTV:

- resolution
- frame rate
- chroma subsampling
- bit depth
- colour space
- white ref is D65
 - ~6500K

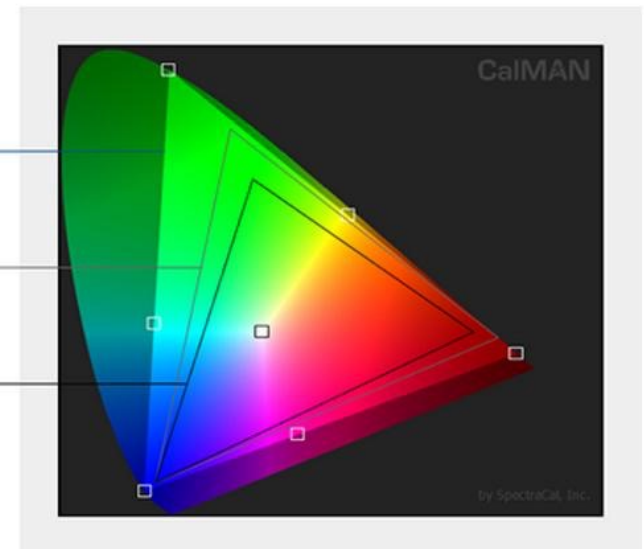


Colorspace Options

ITU-R Rec. 2020 UHD

D-Cinema v1.2 (DCI/P3)

D65, HD Rec.709



<https://www.avforums.com/article/ifa-2014-joe-kane-interview-about-ultra-hd-4k-tv.10702>

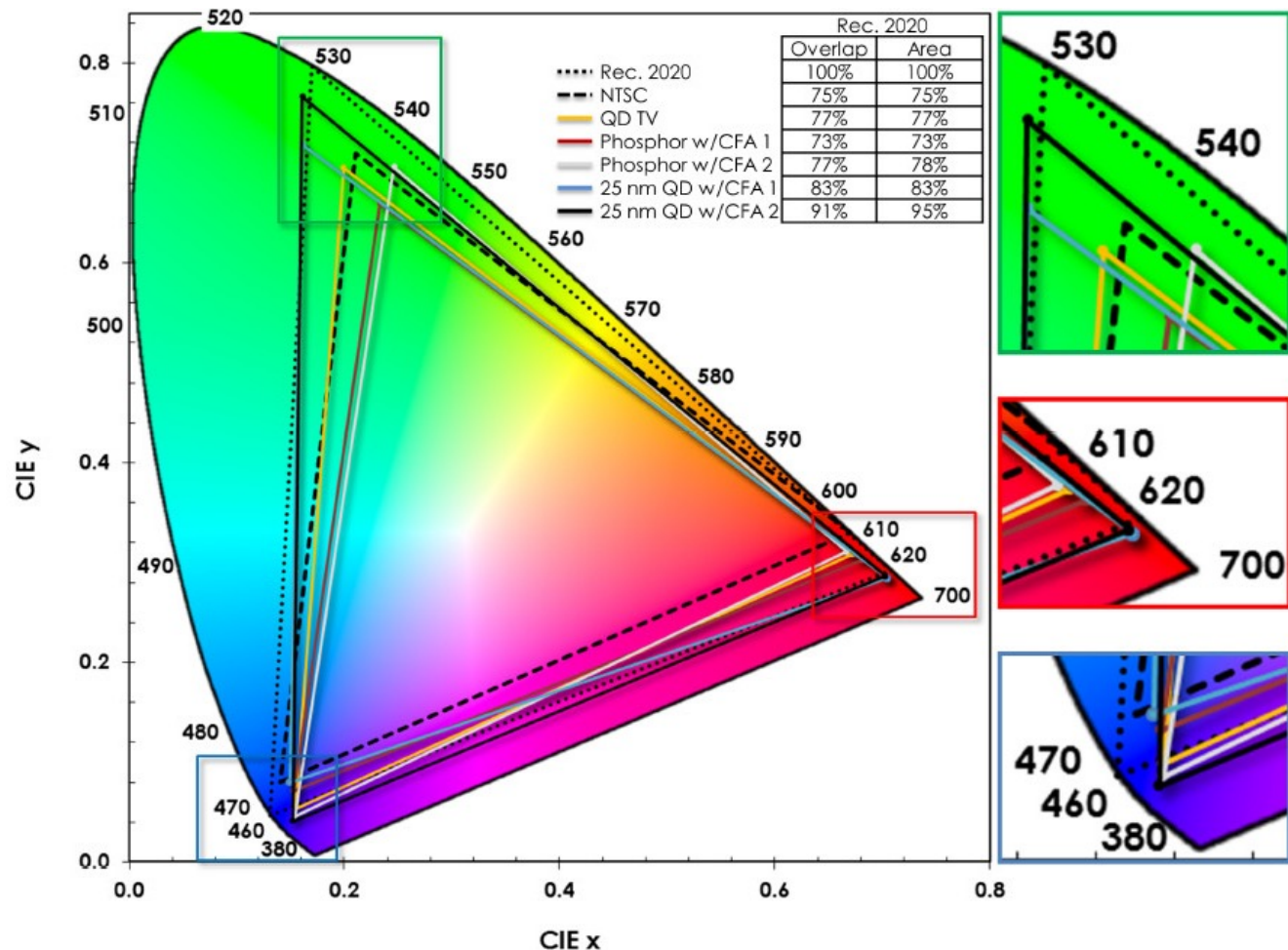
QD / WCG Gamut

WCG = Wide
Colour Gamut

QD: Quantum Dot
Displays

QD – tunable, pure
colour

- very narrow emission spectra
- tuned to wavelength by quantum dot size
- better control of the colour gamut than phosphor LEDs



Phosphor and QD primaries plotted against
Rec.2020 and NTSC reference gamuts

<http://www.display-central.com/subscription-news/editorial-categories/flat-panel/qd-vision-3m-find-advantages-in-quantum-dots-not-surprisingly/>

Additional Slides



Colour Encoding

QAM subcarrier carrying the chrominance
added to the luminance video signal > composite video baseband signal
freq subcarrier =
4.43361875 MHz for PAL, NTSC 4.43
3.579545 MHz for NTSC 3.58

SECAM: FM on two line alternate colour subcarriers 4.25000 and 4.40625 MHz

Phase Alternating Line

phase of part of the colour information on the video signal is reversed with each line
automatically corrects phase errors in the transmission of the signal by cancelling them out
at the expense of vertical frame colour resolution

receivers use a chrominance analog delay line

- stores the received colour information on each line of display
- an average of the colour information from the previous line and the current line is then used to drive the picture tube
- effect – phase errors result in saturation changes – less objectionable than hue changes of NTSC
- drawback i– vertical colour resolution is poorer than NTSC
- human eye colour resolution lower than brightness resolution – PAL effect is not visible
- also – chroma subsampling used

Colour Encoding

Colour carrier

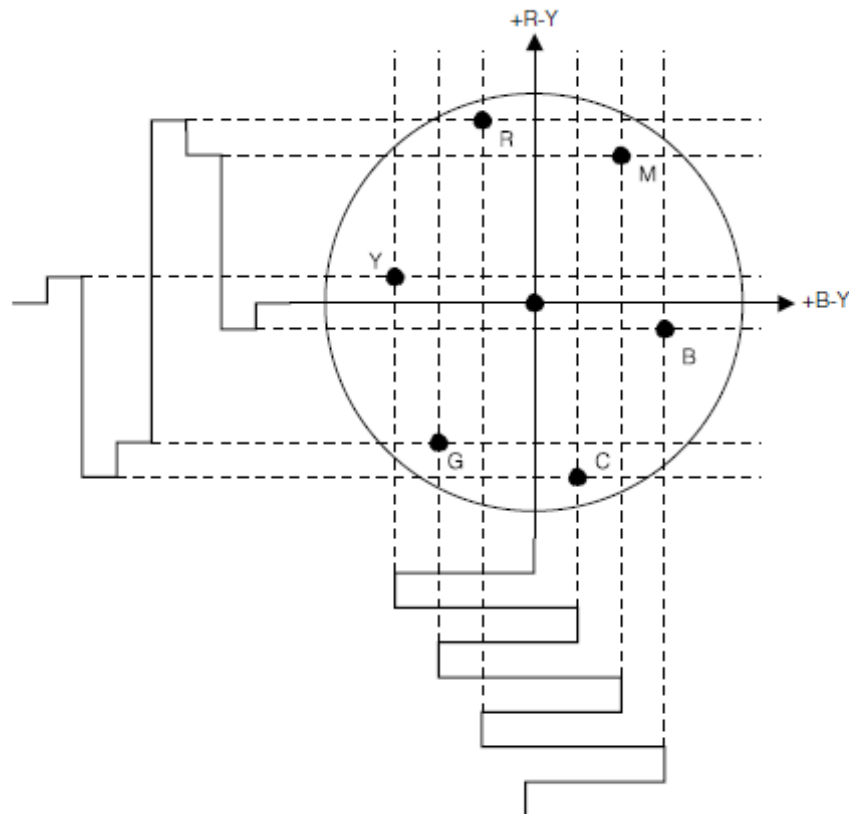
- 4.43361875 MHz frequency
 - due to 283.75 colour clock cycles per line
 - + 25 Hz offset to avoid interferences.
-
- line frequency = 15625 Hz ($625 \text{ lines} \times 50 \text{ Hz} \div 2$)
 - colour carrier frequency:
 - $4.43361875 \text{ MHz} = 283.75 \times 15625 \text{ Hz} + 25 \text{ Hz}.$
-
- original colour carrier is required by the colour decoder to recreate the colour difference signals
 - Since the carrier is not transmitted with the video information it has to be generated locally in the receiver.
 - phase needs to match the transmitted information
 - so 10 cycle burst of colour subcarrier is added to the video signal shortly after the line sync pulse ... colour burst
 - not in phase with the original colour subcarrier:
 - leads by 45 degrees on the odd lines
 - lags it by 45 degrees on the even lines
 - ... this enables the colour decoder circuitry to distinguish the phase of the [R-Y] vector which reverses every line.

Staircase Test Signal

- descending staircase
- peak white >> black
- luminance is unipolar ... no concept of negative brightness
- luminance signal subtracted from the red and blue components
- differences .. these signals are bipolar

Vectorscope

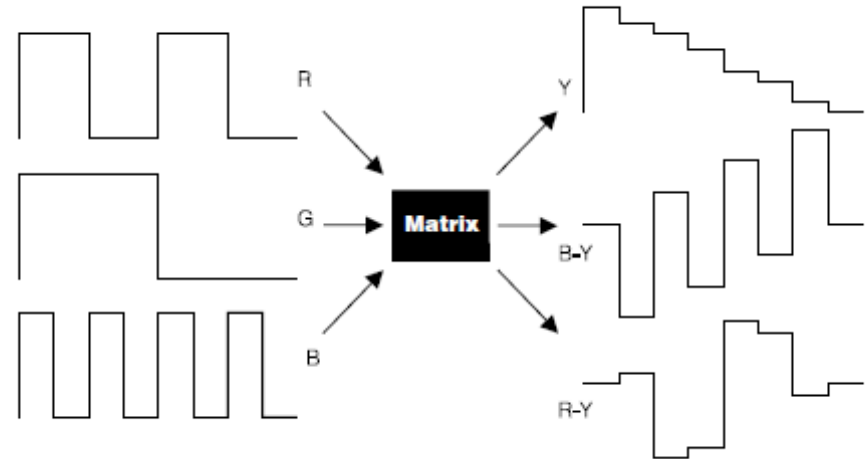
- B-Y causes horizontal deflection
 - R-Y causes vertical deflection
 - ... display having six peripheral dots and two central dots.
-
- central dots
 - white and black bars – not colours – colour difference signals are both zero



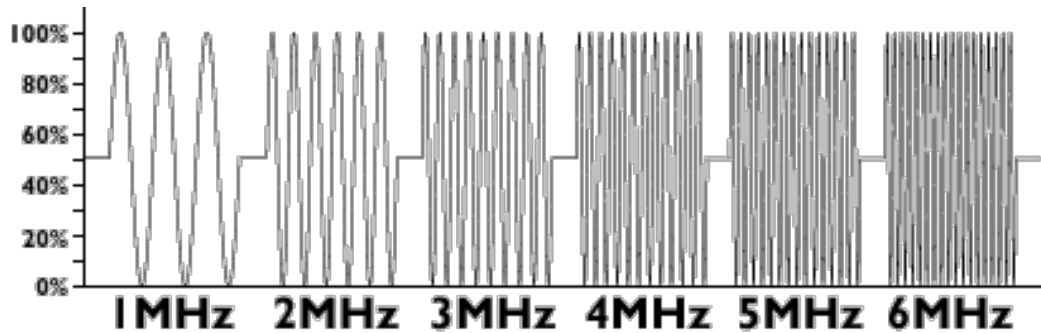
Quadrature Modulation

Colour bar test signal is generated

- RGB square waves of identical amplitude are produced
 - one cycle fits in the active line of the green signal
 - two cycles fit in the active line of the red signal
 - four cycles fit in the active line of the blue signal
-
- eye – non-uniform response to colours
 - R, G and B components are weighted before being added



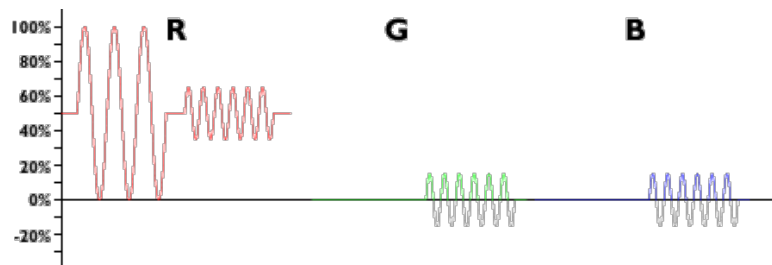
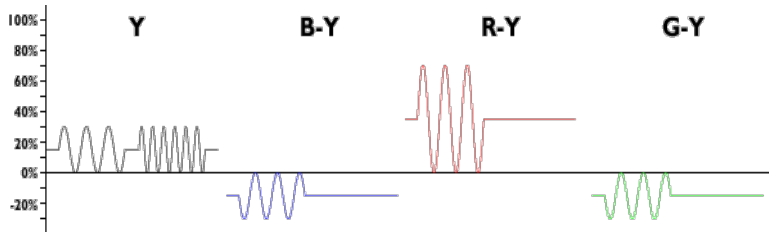
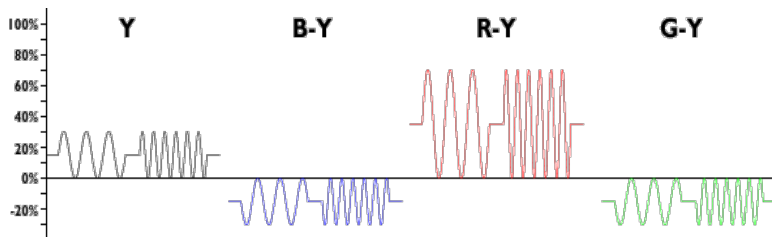
PAL Colour Encoding



Waveform of a multiburst test signal

- ranges from black (0%) to white (100%) with a mean level of 50%

PAL Colour Encoding



Multiburst entirely in one primary colour

- shown – one grating that is within the chrominance passband followed by one that is outside it
- signal is applied to the red channel only, so the green and blue channels are at black level (0%).

The luminance and the three colour-difference channels

- Adding together the luminance and each of the colour difference signals in turn will give the same R, G and B signals as above.

However, the chrominance channels are low-pass filtered

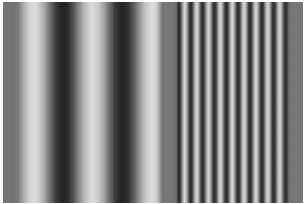
- The lower frequency grating remains unchanged, but the higher one becomes a straight line having the same mean level as the grating.

When the luminance signal is added to each colour-difference signal, the amplitude of the high-frequency grating is reduced in the active channel

- phantom gratings appear in the other two channels, reducing the saturation (and fortuitously increasing the brightness slightly) of the details.
- The red and green channels contain negative-going information which will either be clipped by the display circuitry or cut off in the TV

PAL Colour Encoding

The loss of luminance resolution is less pronounced in areas of saturated green and more pronounced in saturated blue because of the relative contributions made by the colour separation signals to the luminance signal



This sequence of screenshots shows the two sections of multiburst as a black-and-white signal...



..as a signal in the red channel only as displayed on an RGB monitor before encoding...



..and as displayed after decoding.